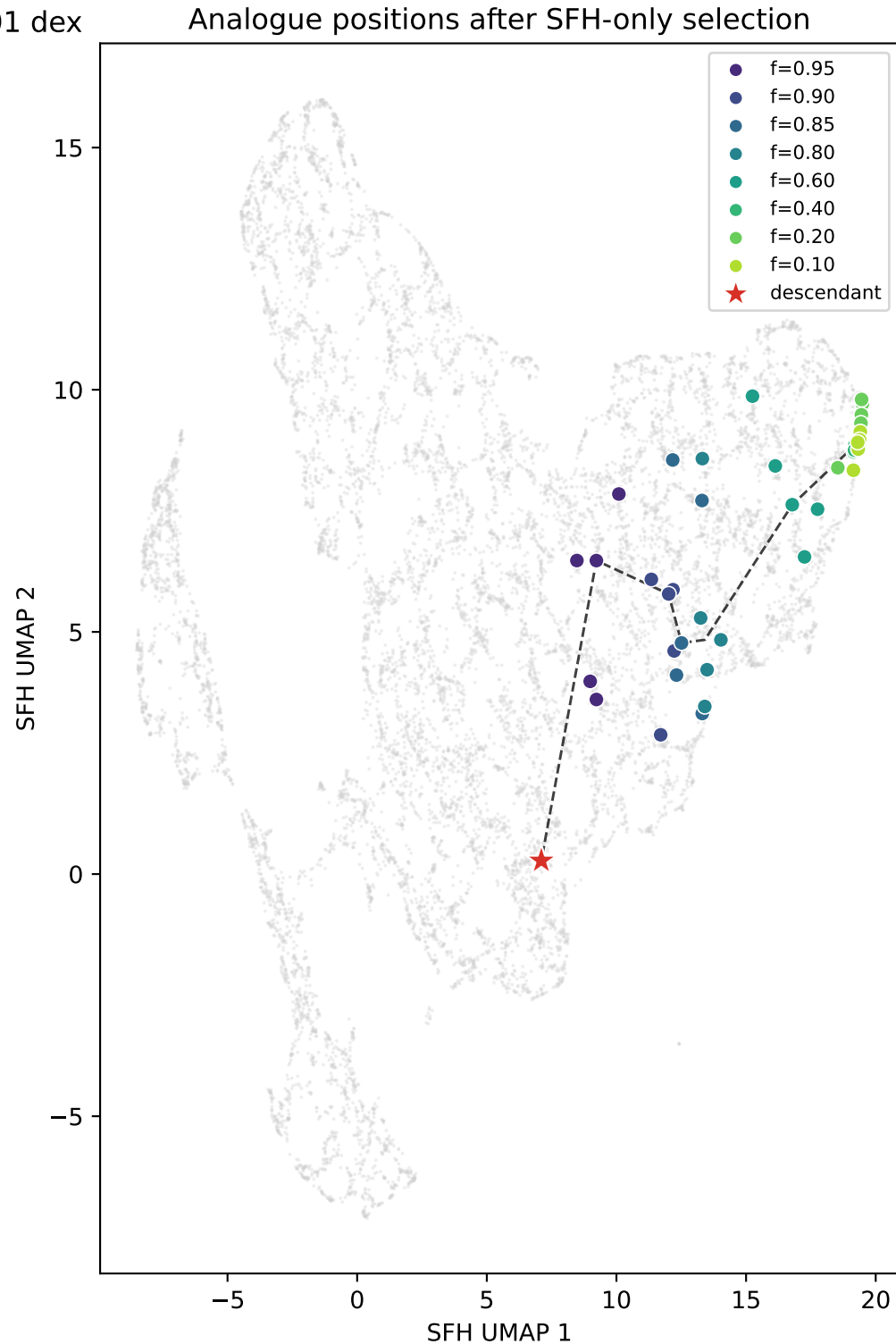
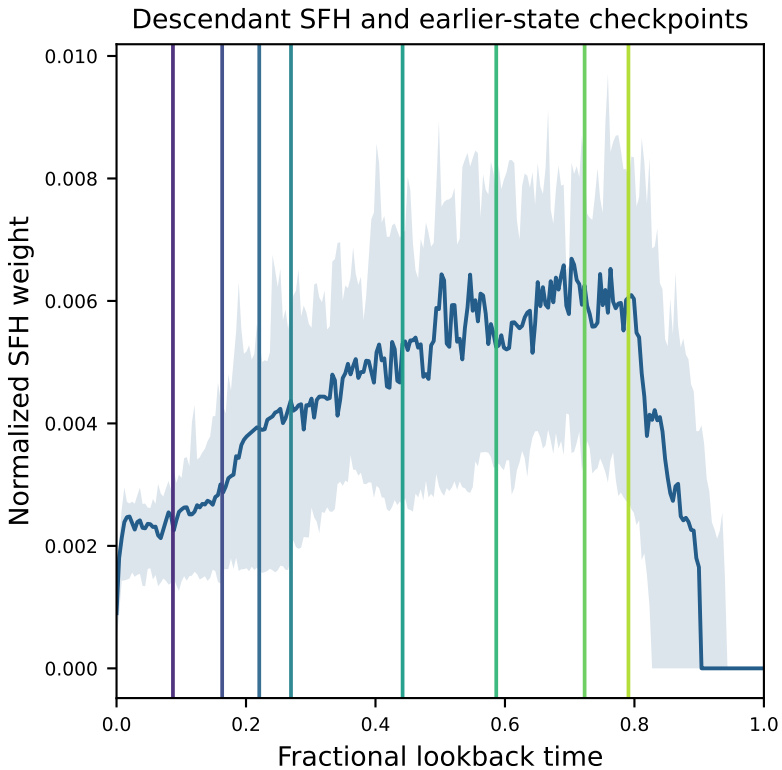
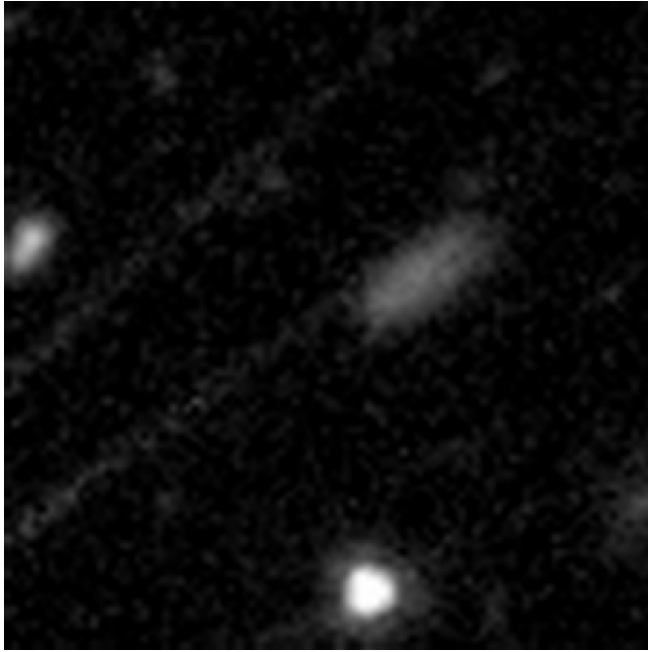
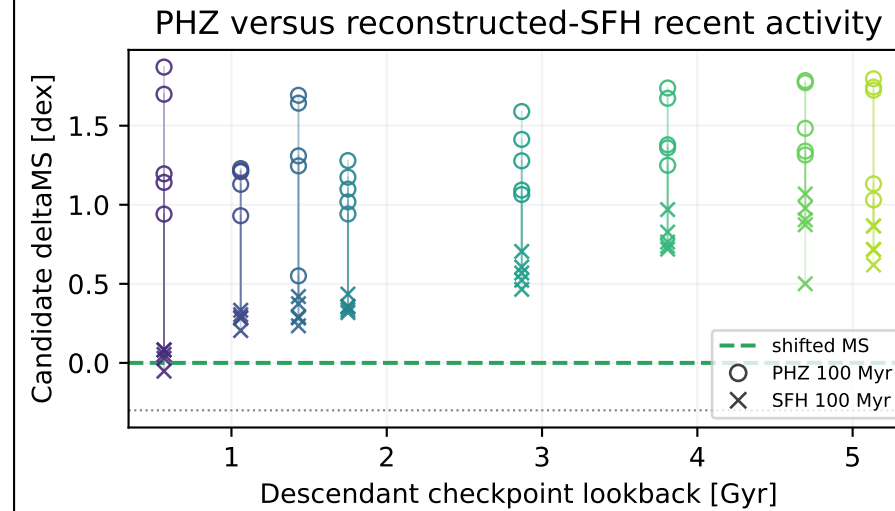


Pilot SFH-selected progenitor-analogue sequence

Descendant 2736881428684070683
 log M*=10.22, z=0.830, log sSFR=-8.97
 PHZ deltaMS=+1.12; SFH-100Myr deltaMS=-0.01 dex



f	L	logM	T	N	zmed
0.95	0.57	10.20	2.00	2,001	0.53
0.90	1.06	10.17	2.00	1,988	0.53
0.85	1.43	10.15	2.00	2,007	0.52
0.80	1.75	10.12	2.00	2,029	0.52
0.60	2.87	10.00	2.00	2,039	0.47
0.40	3.81	9.82	2.00	1,955	0.43
0.20	4.69	9.52	1.80	1,687	0.35
0.10	5.13	9.22	1.36	1,329	0.29



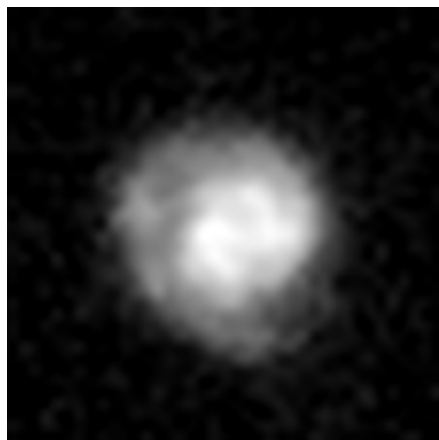
Selection variables

- predicted stellar mass, within +/-0.15 dex
- cumulative SFH before each descendant state
- 2 Gyr physical-time comparison when available

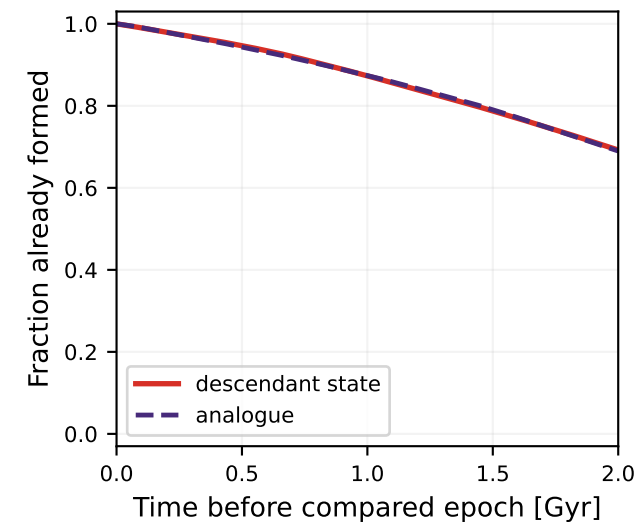
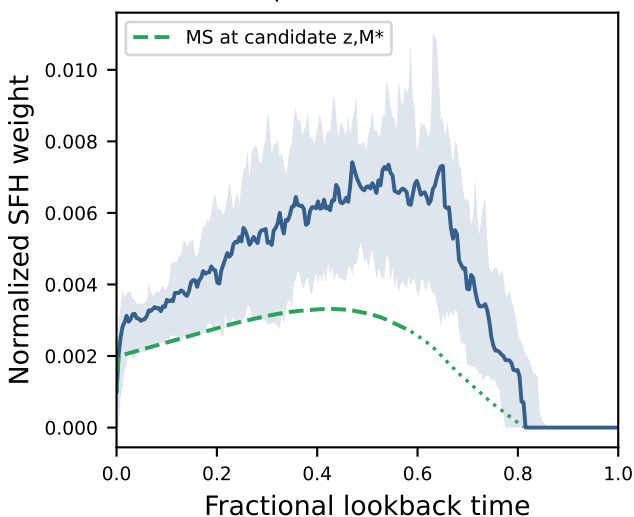
Candidate redshift, morphology, deltaMS, image embedding, and UMAP position are outcomes, not selection variables. The dashed path joins ensemble medians, not one galaxy's orbit.

Earlier state $f=0.95$: lookback=0.57 Gyr, predicted $\log M^*=10.20$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

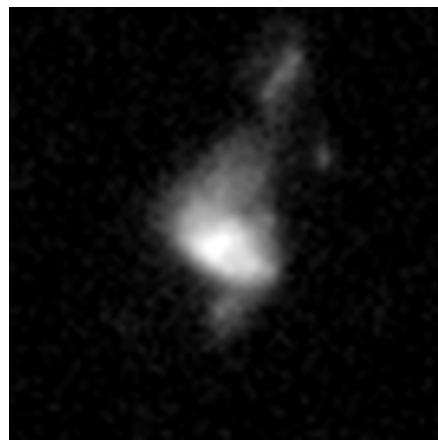
rank 1: 2745221494645136570
 $\log M^*=10.31$ (+0.11) $z=0.822$
 PHZ $\Delta MS=+0.94$; SFH-100Myr $\Delta MS=+0.08$
 $D=0.002$



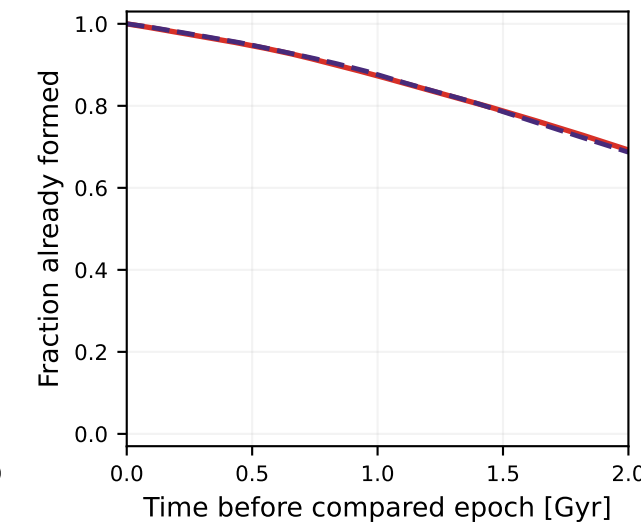
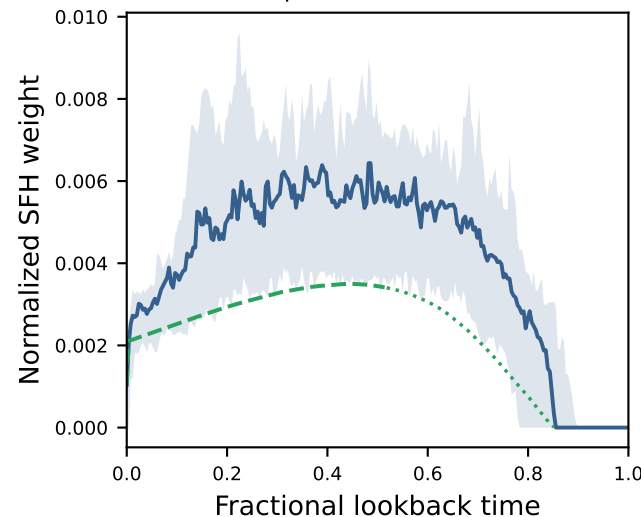
Complete candidate SFH



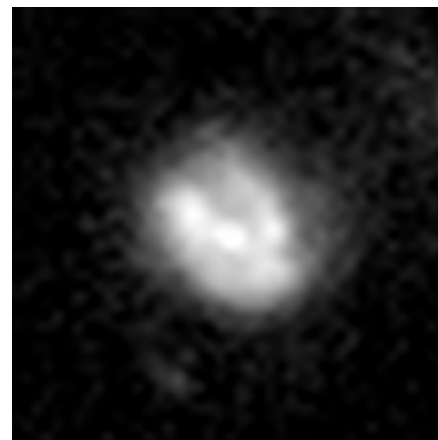
rank 2: 2684186475660615732
 $\log M^*=10.15$ (-0.05) $z=0.754$
 PHZ $\Delta MS=+1.14$; SFH-100Myr $\Delta MS=+0.05$
 $D=0.002$



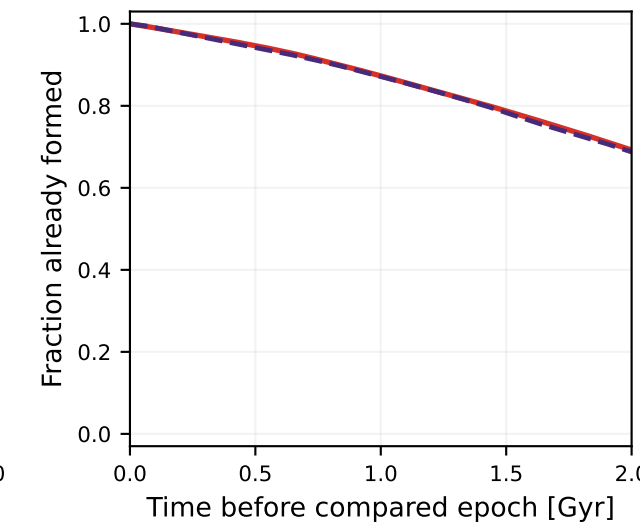
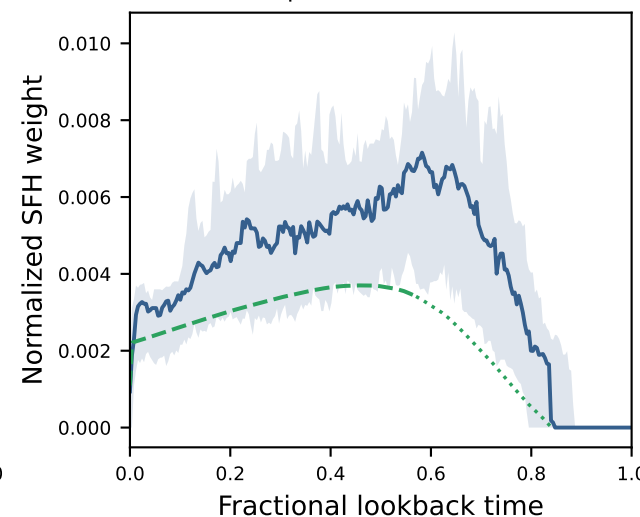
Complete candidate SFH



rank 3: 2649613390679878335
 $\log M^*=10.20$ (-0.00) $z=0.838$
 PHZ $\Delta MS=+1.20$; SFH-100Myr $\Delta MS=+0.03$
 $D=0.003$



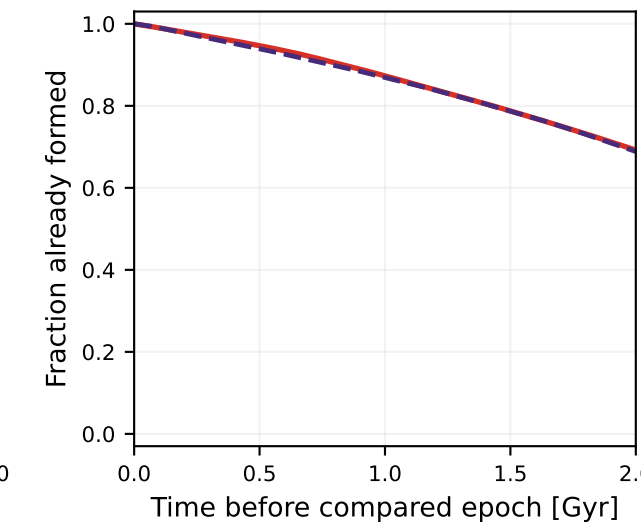
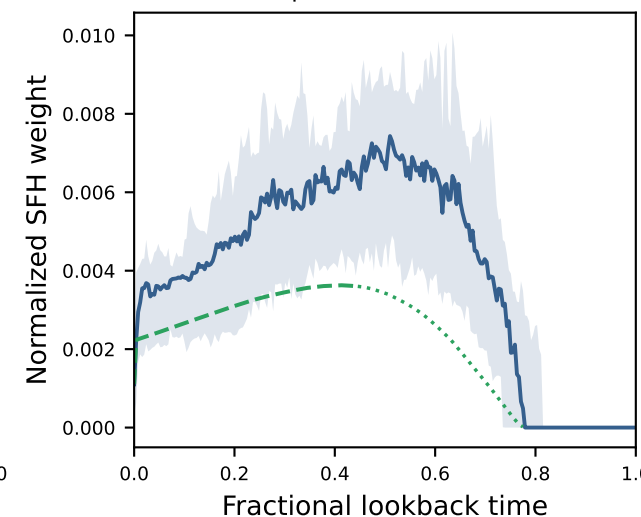
Complete candidate SFH



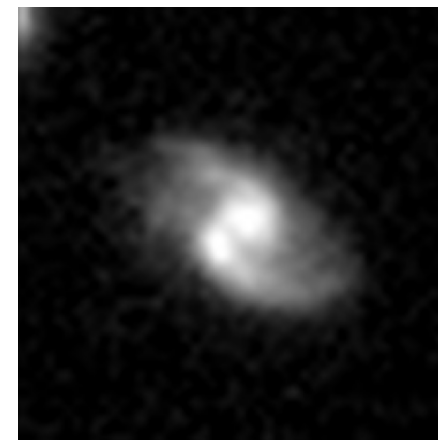
rank 4: 2729657949640354142
 $\log M^*=10.05$ (-0.15) $z=0.738$
 PHZ $\Delta MS=+1.87$; SFH-100Myr $\Delta MS=+0.08$
 $D=0.003$



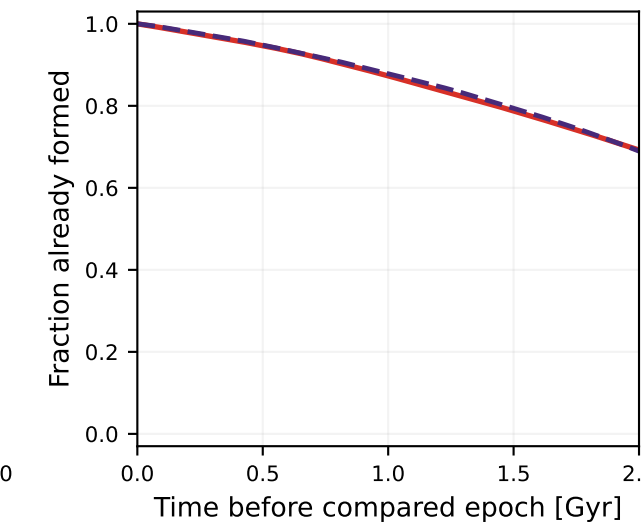
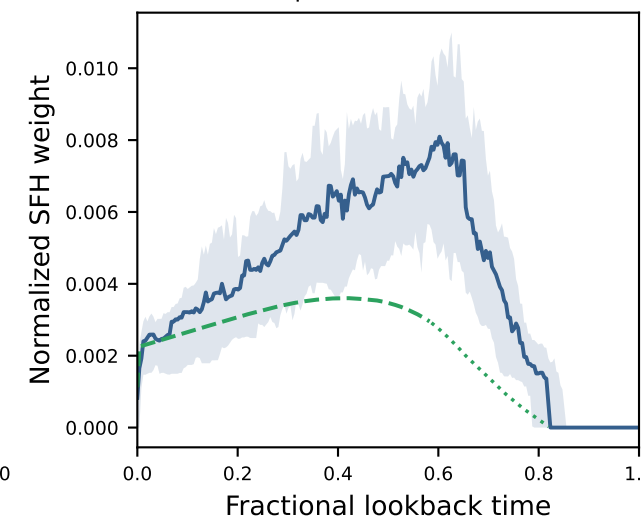
Complete candidate SFH



rank 5: 2644952133675305599
 $\log M^*=10.28$ (+0.08) $z=0.928$
 PHZ $\Delta MS=+1.70$; SFH-100Myr $\Delta MS=-0.0$
 $D=0.004$



Complete candidate SFH

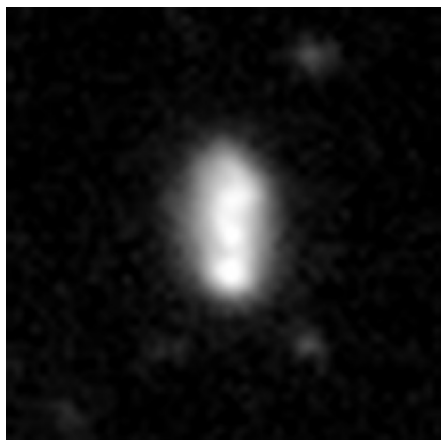
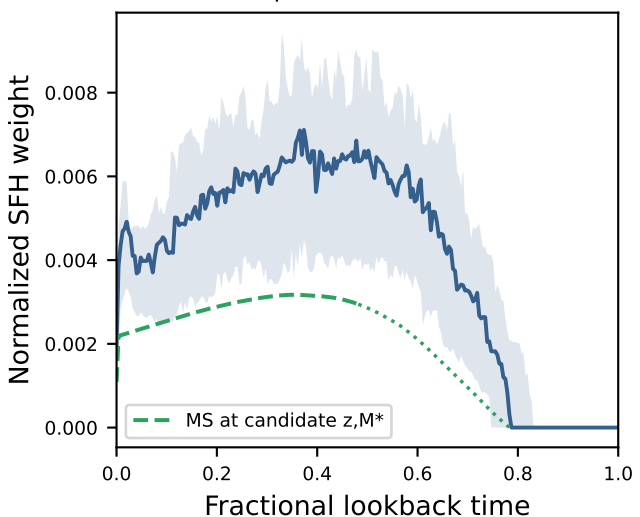


Earlier state $f=0.90$: lookback=1.06 Gyr, predicted $\log M^*=10.17$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

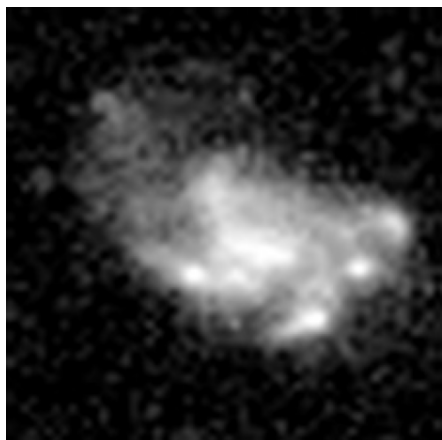
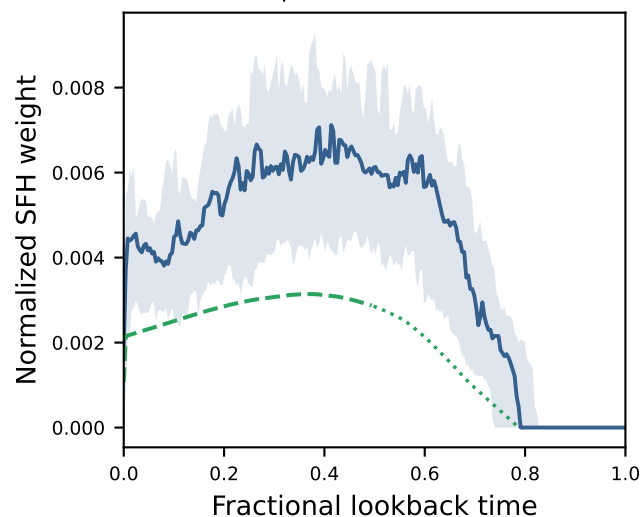
rank 1: 2674050761656456059 log $M^*=10.17$ (-0.00) $z=0.812$ PHZ $\Delta MS=+1.22$; SFH-100Myr $\Delta MS=+0.29$ D=0.003	rank 2: 2653869952671053010 log $M^*=10.21$ (+0.03) $z=0.820$ PHZ $\Delta MS=+1.23$; SFH-100Myr $\Delta MS=+0.28$ D=0.003	rank 3: 2759585612644997087 log $M^*=10.29$ (+0.12) $z=0.800$ PHZ $\Delta MS=+1.21$; SFH-100Myr $\Delta MS=+0.33$ D=0.003	rank 4: 2705609495650354453 log $M^*=10.18$ (+0.00) $z=0.768$ PHZ $\Delta MS=+0.93$; SFH-100Myr $\Delta MS=+0.20$ D=0.004	rank 5: 2686104951674420136 log $M^*=10.22$ (+0.05) $z=0.808$ PHZ $\Delta MS=+1.13$; SFH-100Myr $\Delta MS=+0.0$ D=0.004
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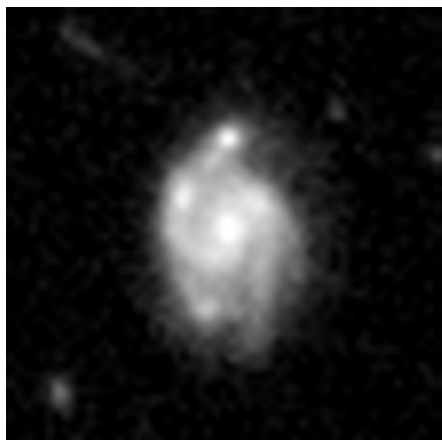
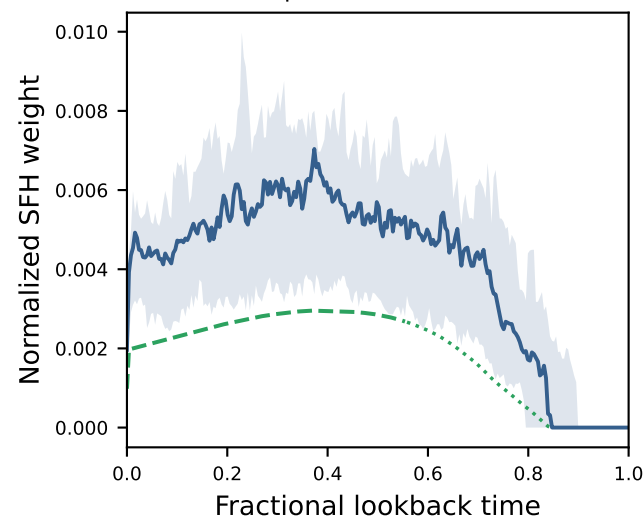
Complete candidate SFH



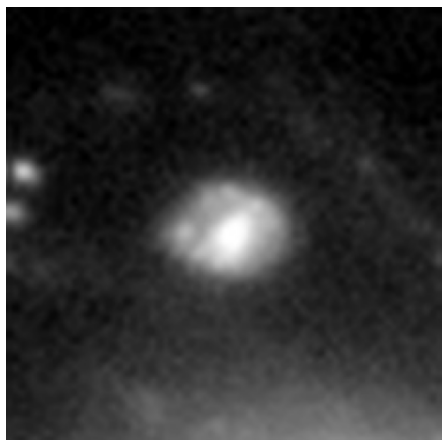
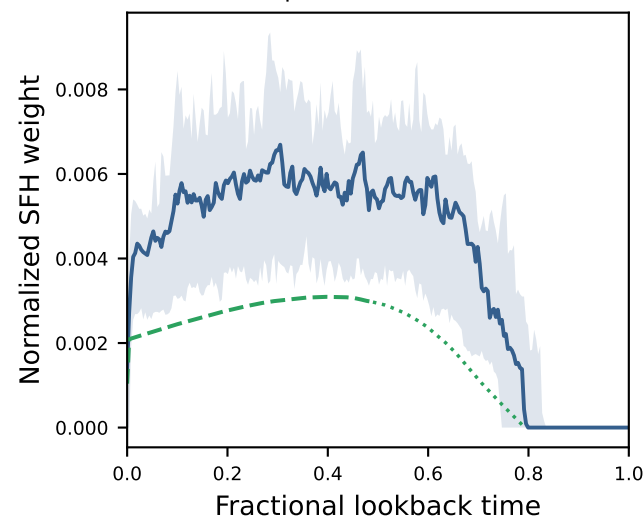
Complete candidate SFH



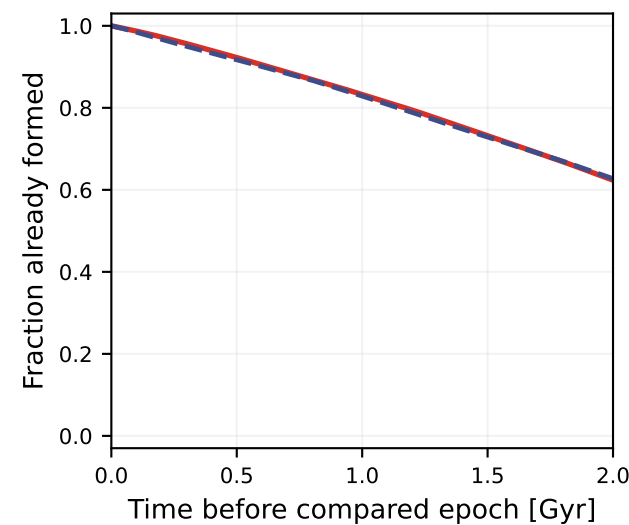
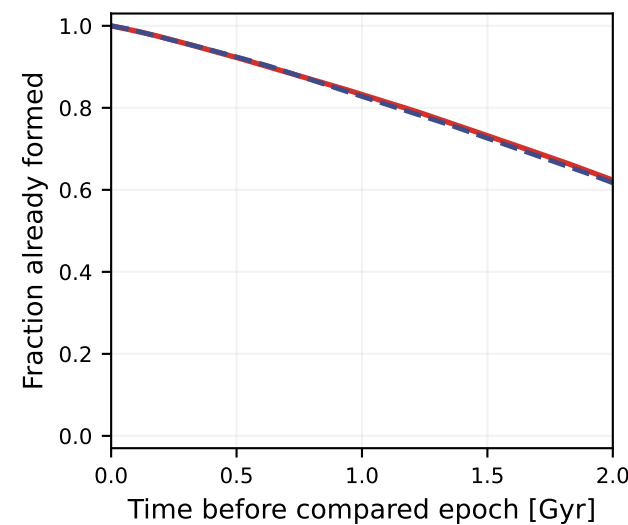
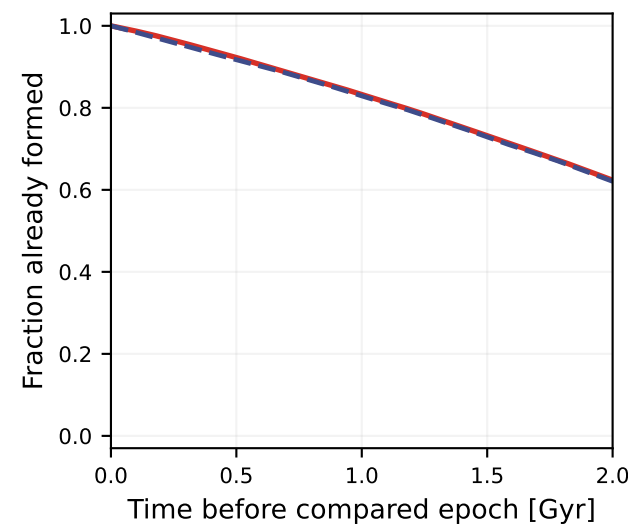
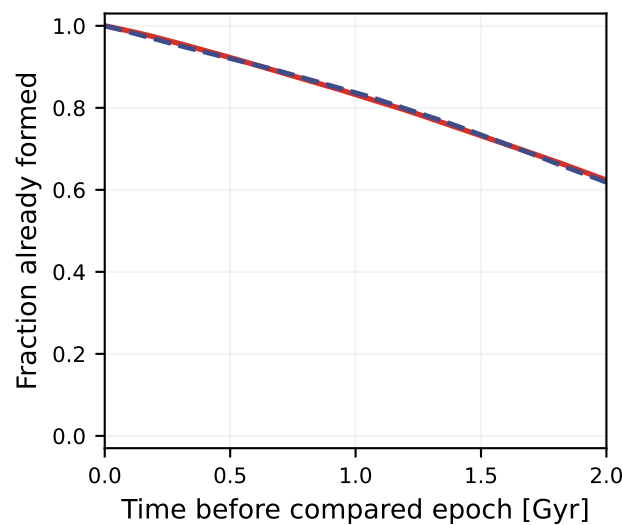
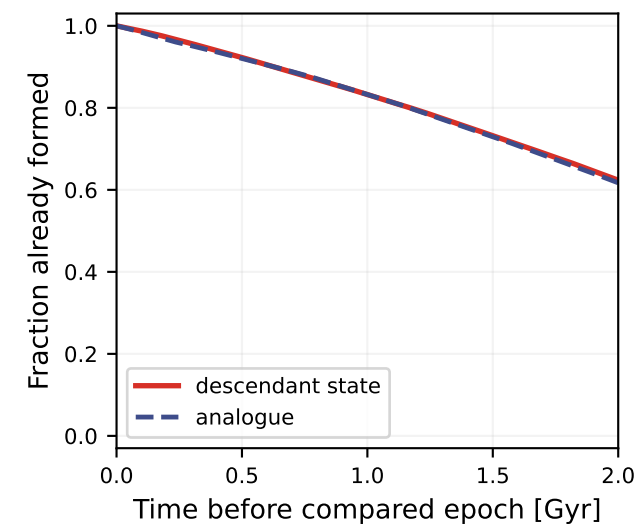
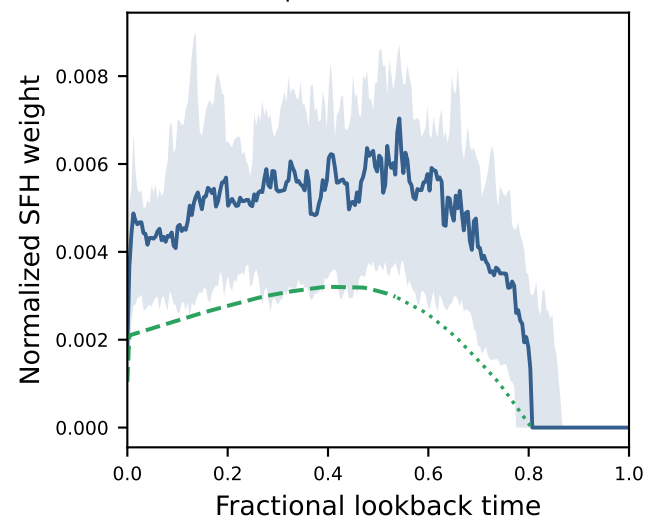
Complete candidate SFH



Complete candidate SFH

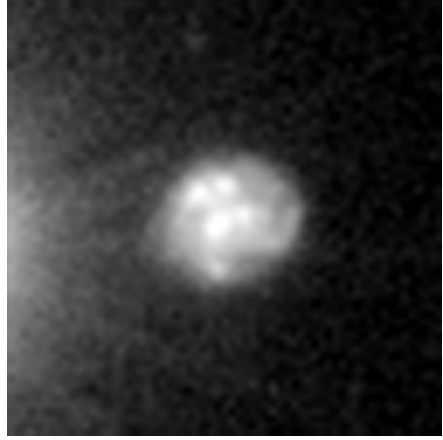


Complete candidate SFH

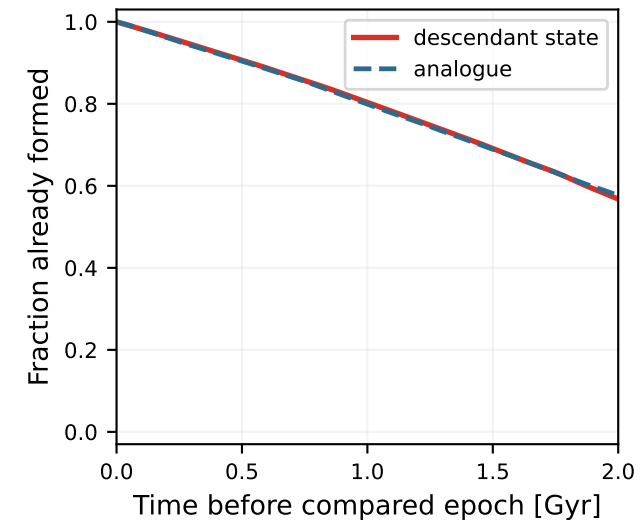
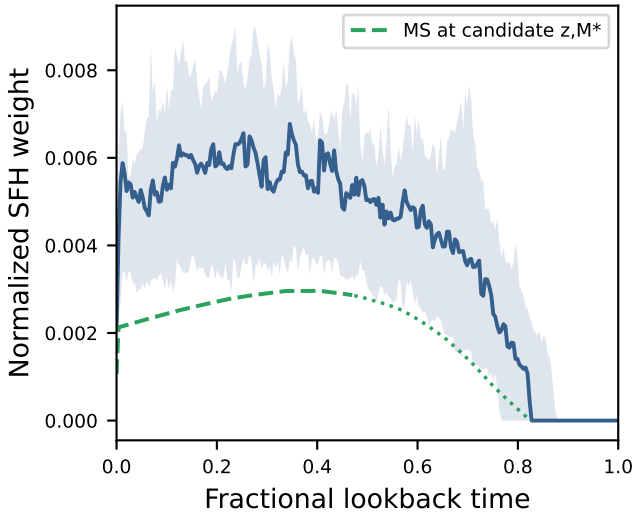


Earlier state $f=0.85$: lookback=1.43 Gyr, predicted $\log M^*=10.15$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

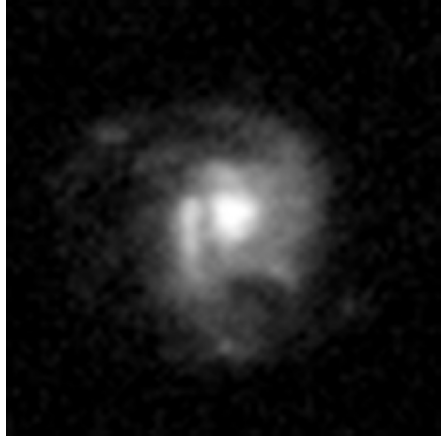
rank 1: 2665175169674165314
 $\log M^*=10.18$ (+0.03) $z=0.790$
 PHZ $\Delta MS=+1.69$; SFH-100Myr $\Delta MS=+0.37$
 $D=0.003$



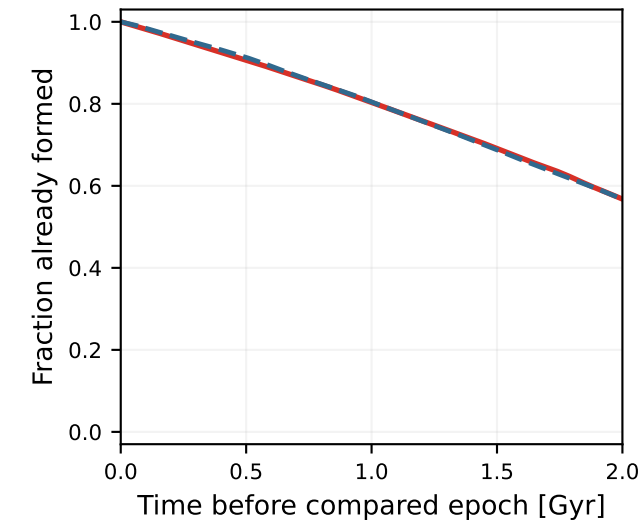
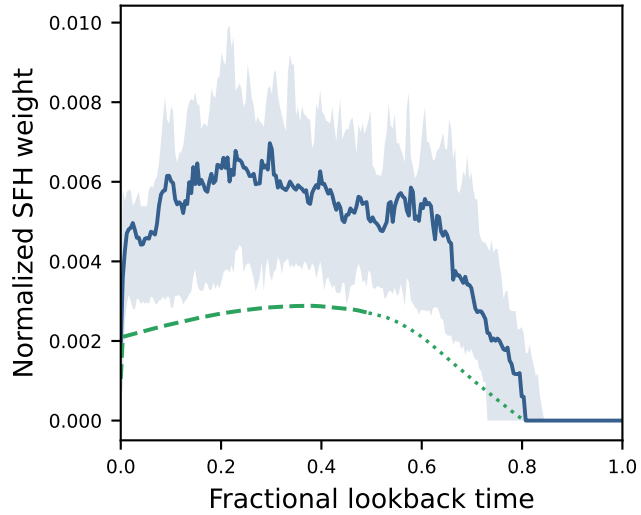
Complete candidate SFH



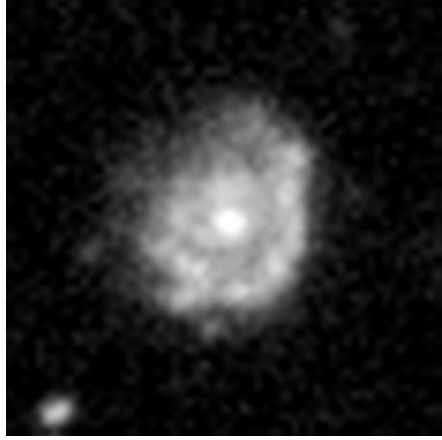
rank 2: 2717907506655294709
 $\log M^*=10.22$ (+0.07) $z=0.810$
 PHZ $\Delta MS=+1.64$; SFH-100Myr $\Delta MS=+0.29$
 $D=0.003$



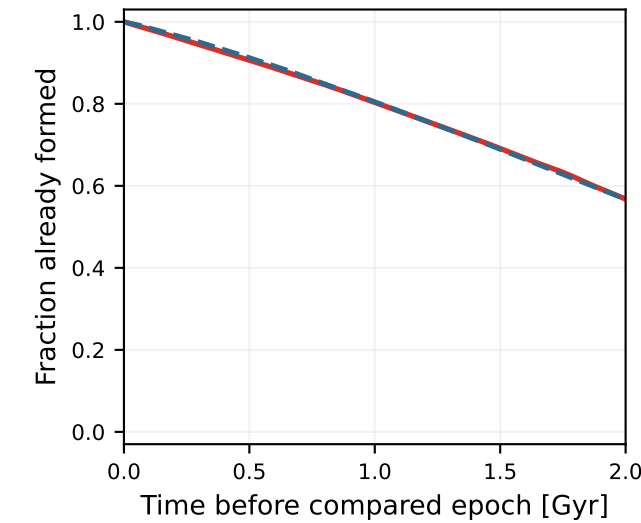
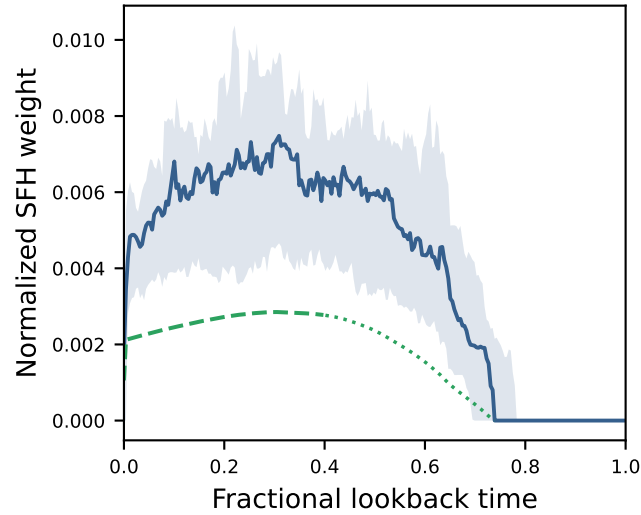
Complete candidate SFH



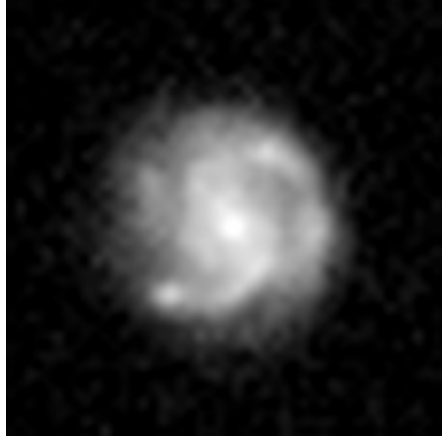
rank 3: 2703655474674948884
 $\log M^*=10.12$ (-0.03) $z=0.744$
 PHZ $\Delta MS=+0.55$; SFH-100Myr $\Delta MS=+0.28$
 $D=0.003$



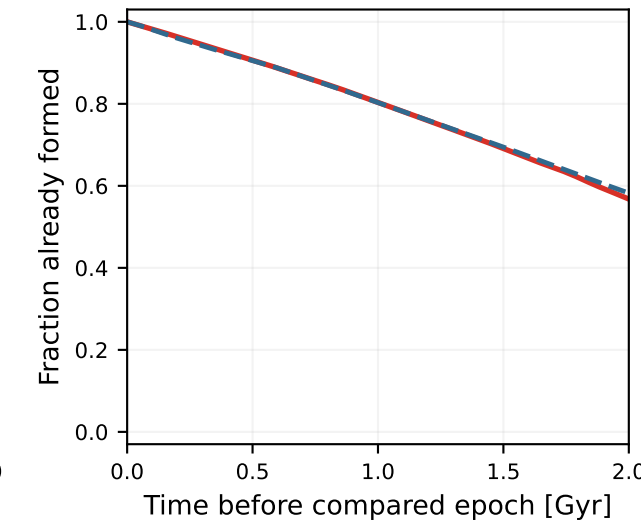
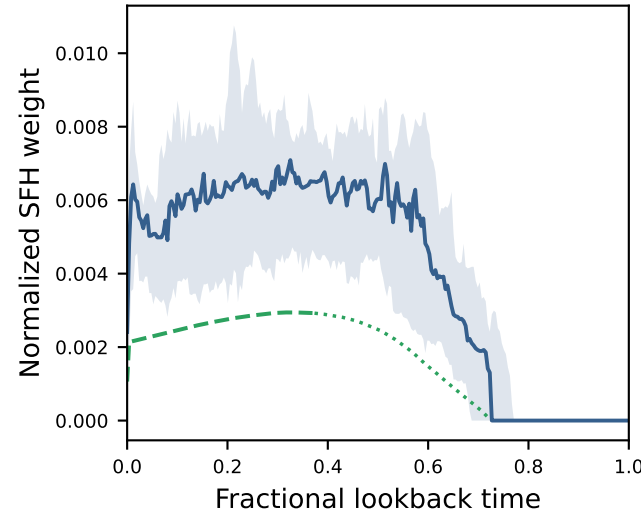
Complete candidate SFH



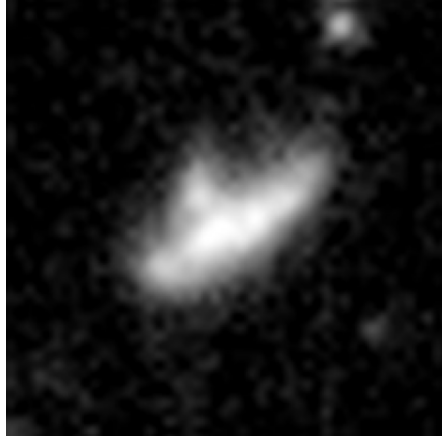
rank 4: 2718255177647134336
 $\log M^*=10.06$ (-0.09) $z=0.710$
 PHZ $\Delta MS=+1.25$; SFH-100Myr $\Delta MS=+0.42$
 $D=0.003$



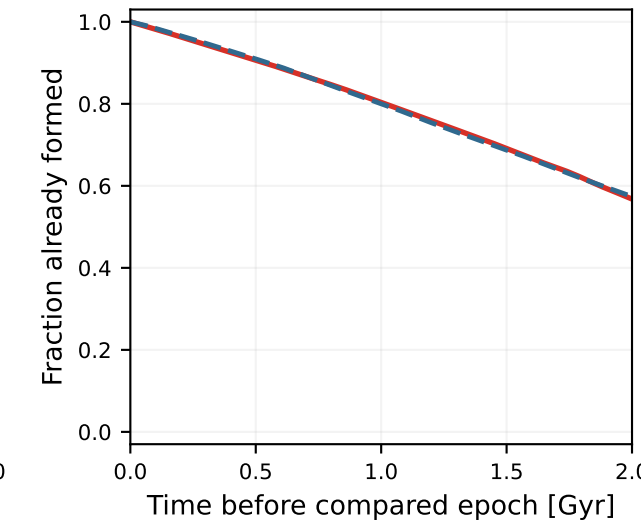
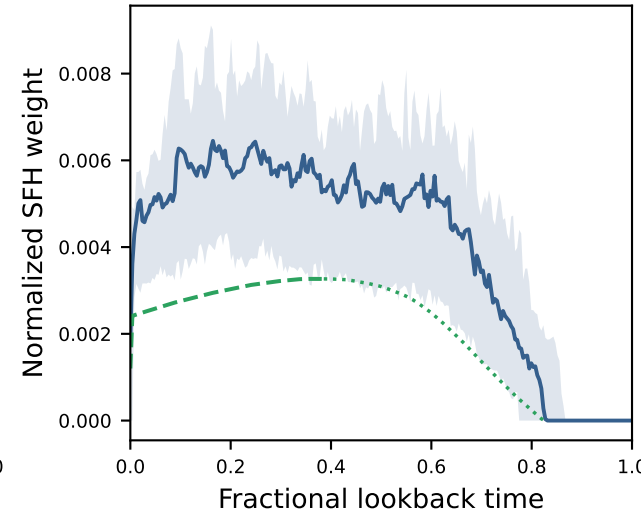
Complete candidate SFH



rank 5: 2631843144671946880
 $\log M^*=10.04$ (-0.11) $z=0.808$
 PHZ $\Delta MS=+1.31$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.003$

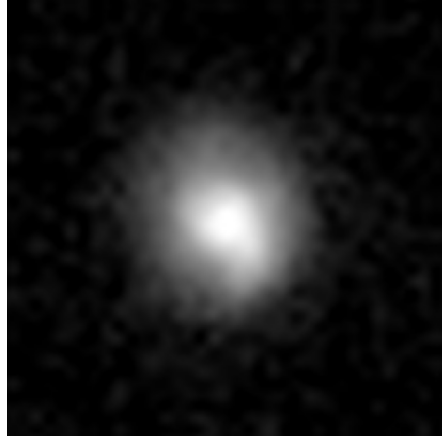


Complete candidate SFH

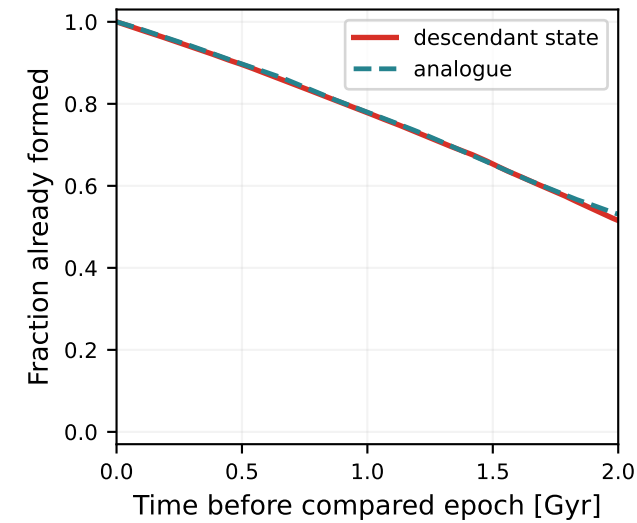
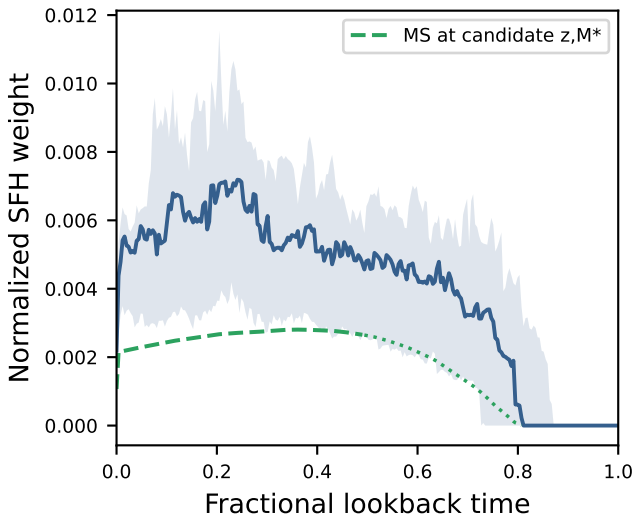


Earlier state $f=0.80$: lookback=1.75 Gyr, predicted $\log M^*=10.12$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

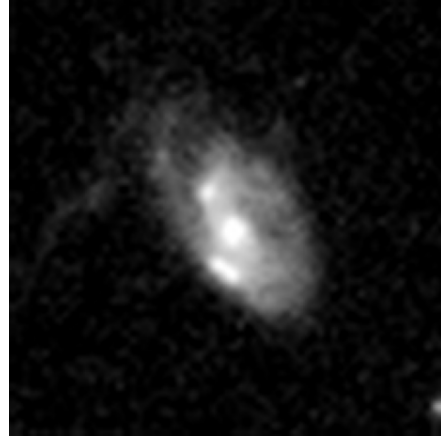
rank 1: 2629833978656767409
 $\log M^*=10.23 (+0.11)$ $z=0.840$
 PHZ $\Delta MS=+1.10$; SFH-100Myr $\Delta MS=+0.35$
 $D=0.003$



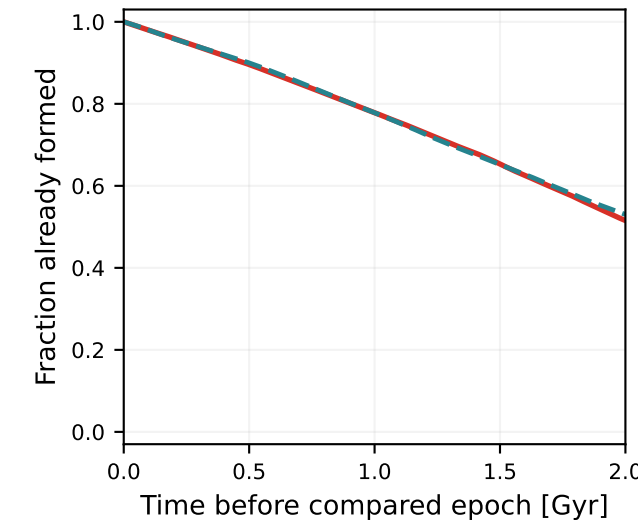
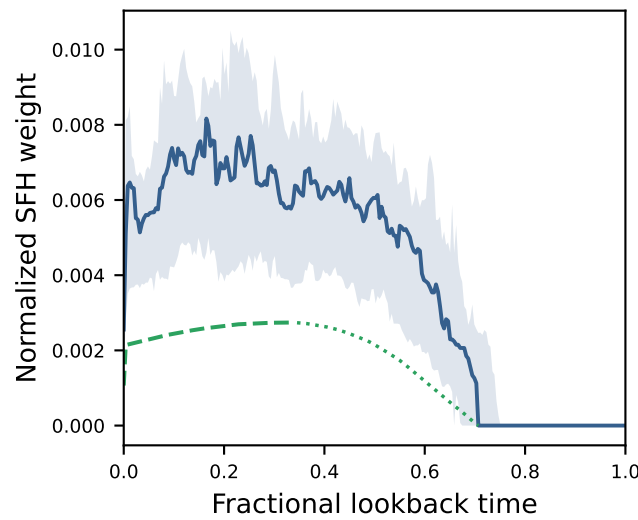
Complete candidate SFH



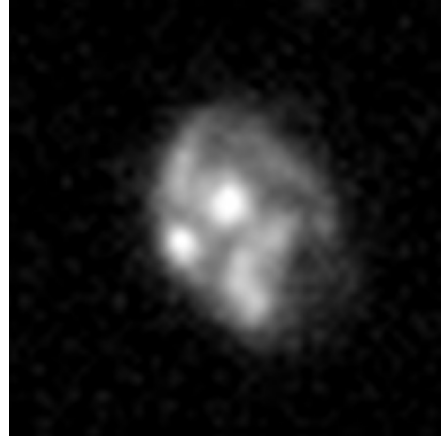
rank 2: 2694980736676355071
 $\log M^*=10.06 (-0.06)$ $z=0.712$
 PHZ $\Delta MS=+1.02$; SFH-100Myr $\Delta MS=+0.44$
 $D=0.003$



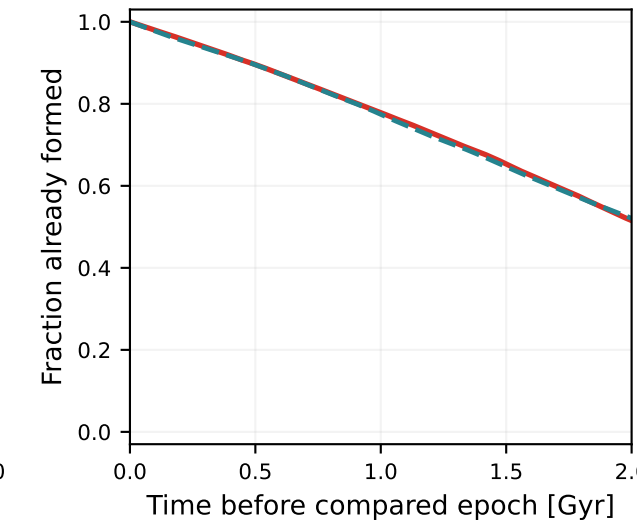
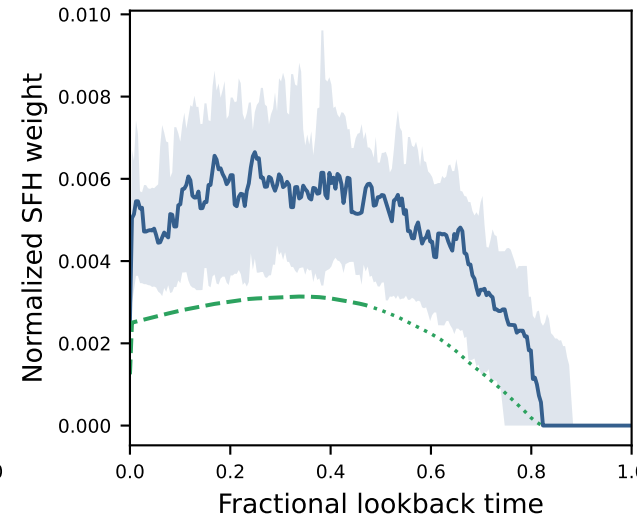
Complete candidate SFH



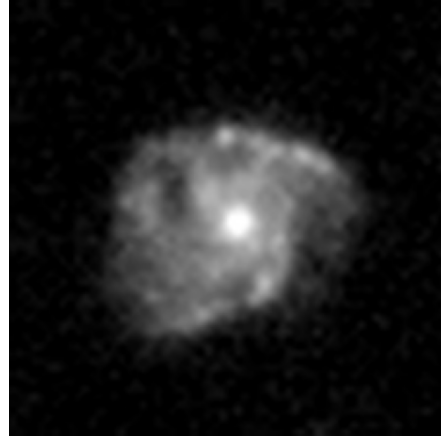
rank 3: 2641625667665171672
 $\log M^*=10.20 (+0.07)$ $z=0.988$
 PHZ $\Delta MS=+1.28$; SFH-100Myr $\Delta MS=+0.32$
 $D=0.003$



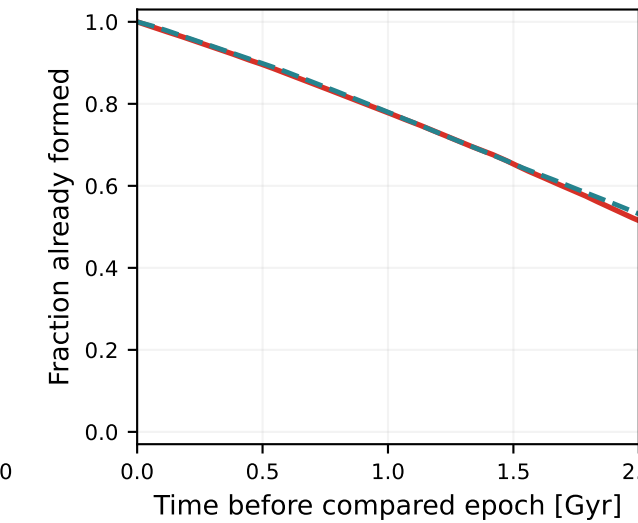
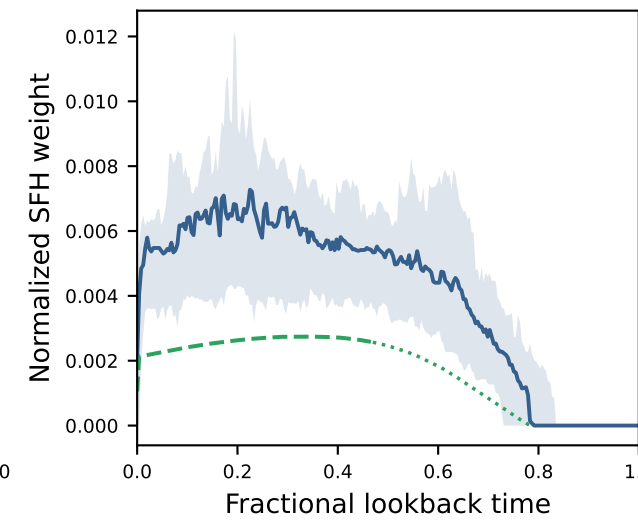
Complete candidate SFH



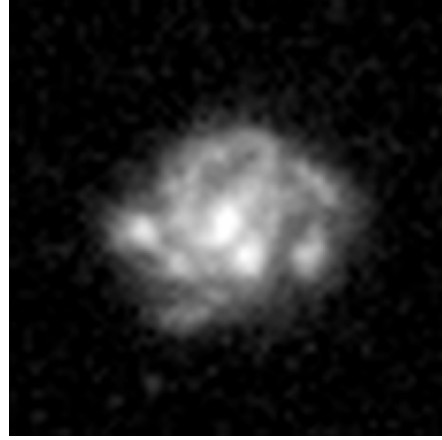
rank 4: 2744605794673035135
 $\log M^*=10.22 (+0.10)$ $z=0.820$
 PHZ $\Delta MS=+0.94$; SFH-100Myr $\Delta MS=+0.33$
 $D=0.004$



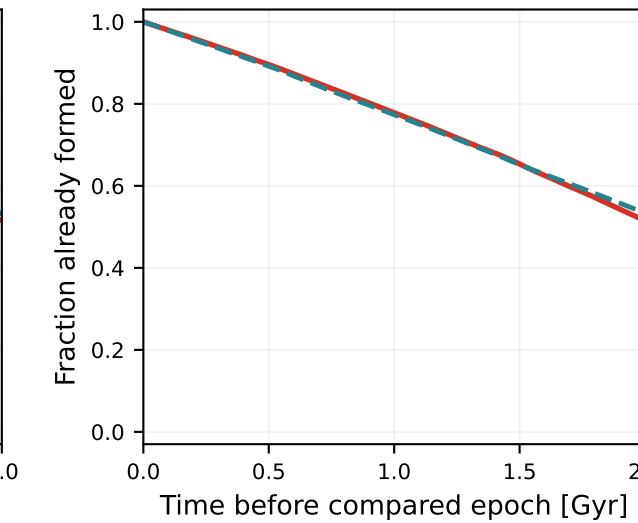
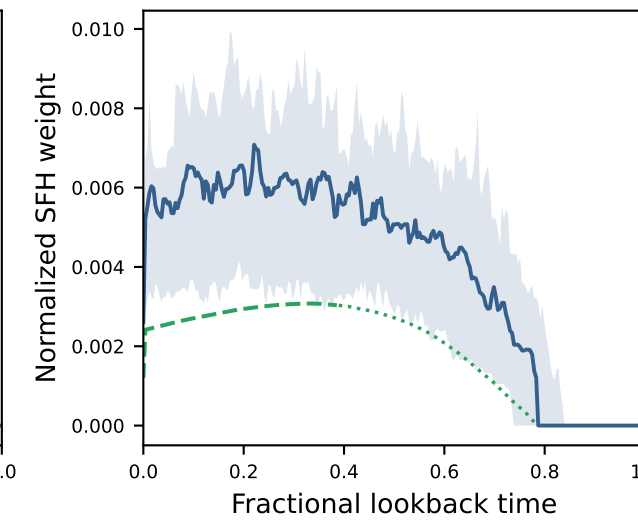
Complete candidate SFH



rank 5: 2699954644647219081
 $\log M^*=10.08 (-0.05)$ $z=0.840$
 PHZ $\Delta MS=+1.17$; SFH-100Myr $\Delta MS=+0.40$
 $D=0.005$

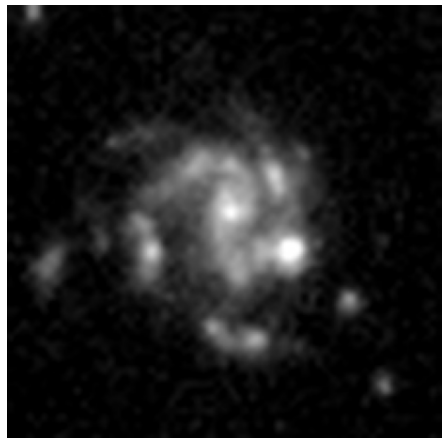


Complete candidate SFH

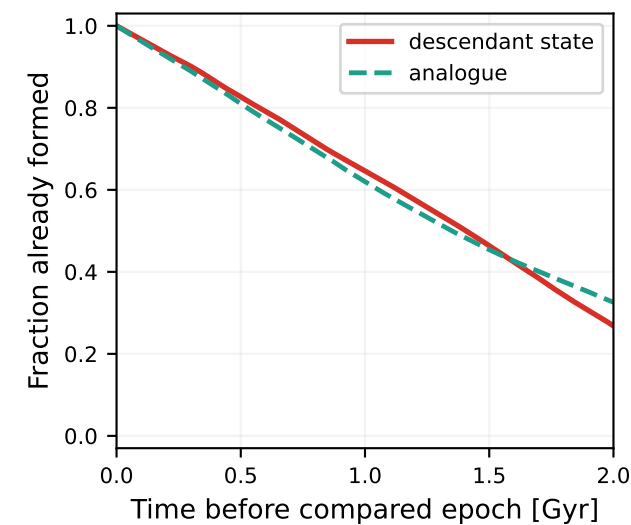
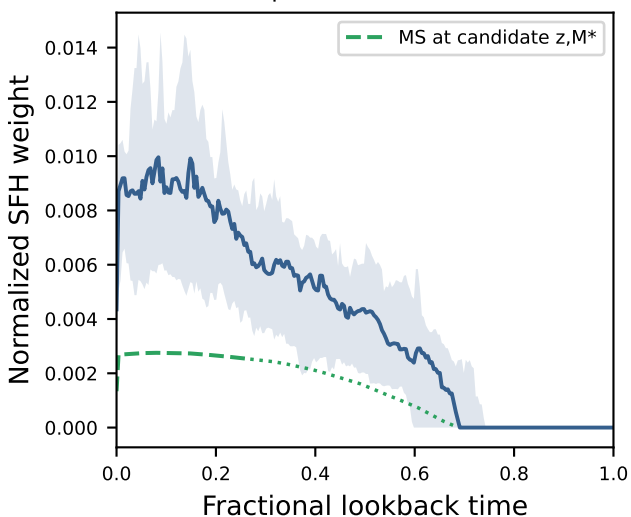


Earlier state $f=0.60$: lookback=2.87 Gyr, predicted $\log M^*=10.00$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

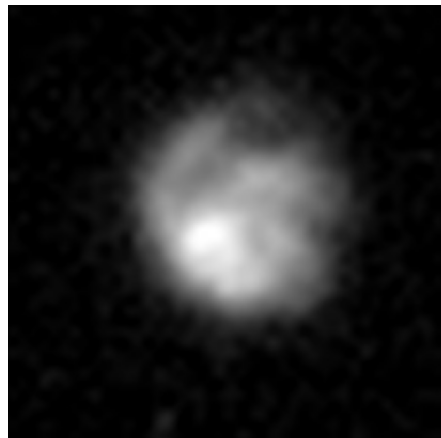
rank 1: 2689545645661527879
 $\log M^*=10.07 (+0.07)$ $z=0.964$
 PHZ $\Delta MS=+1.41$; SFH-100Myr $\Delta MS=+0.52$
 $D=0.020$



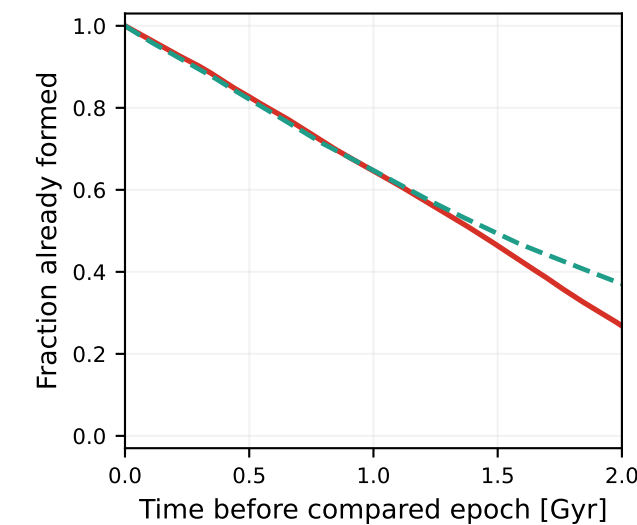
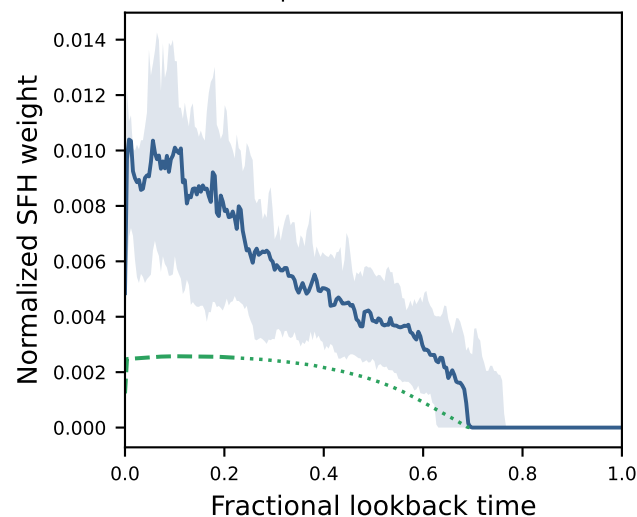
Complete candidate SFH



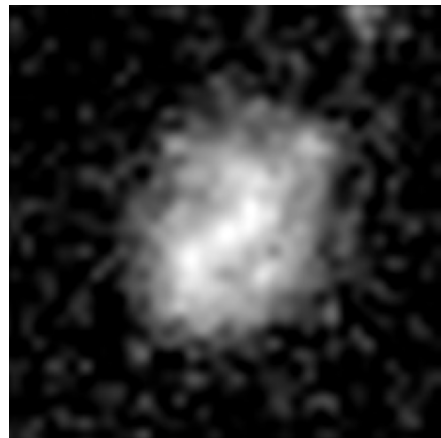
rank 2: 2754970180654731499
 $\log M^*=10.01 (+0.01)$ $z=0.816$
 PHZ $\Delta MS=+1.09$; SFH-100Myr $\Delta MS=+0.61$
 $D=0.021$



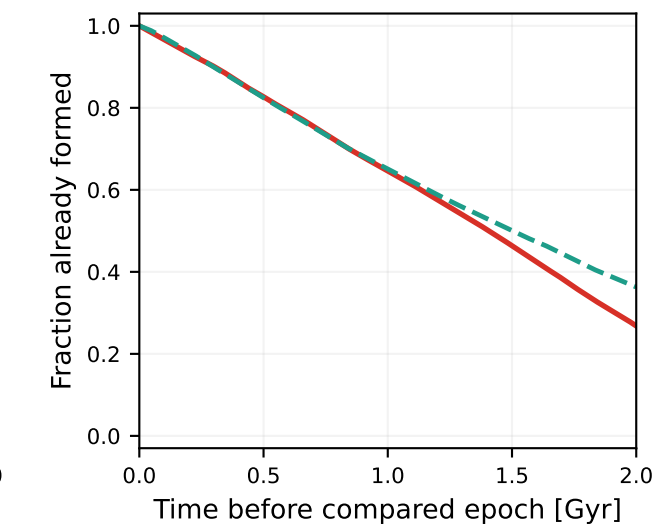
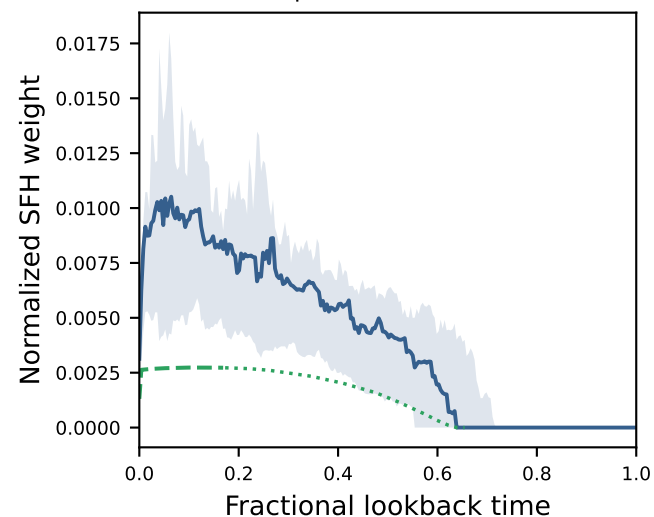
Complete candidate SFH



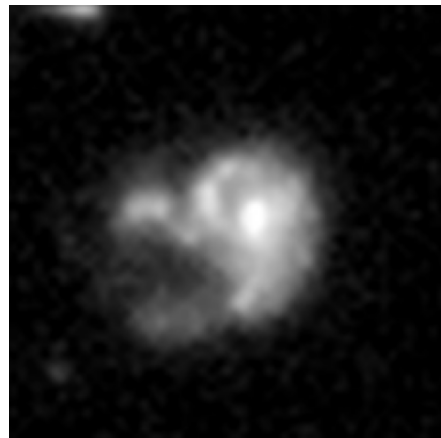
rank 3: 2721420199689695654
 $\log M^*=9.92 (-0.08)$ $z=0.802$
 PHZ $\Delta MS=+1.28$; SFH-100Myr $\Delta MS=+0.47$
 $D=0.022$



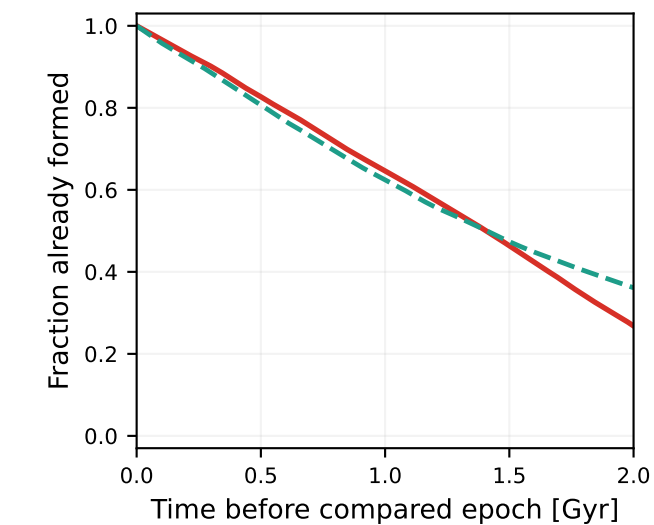
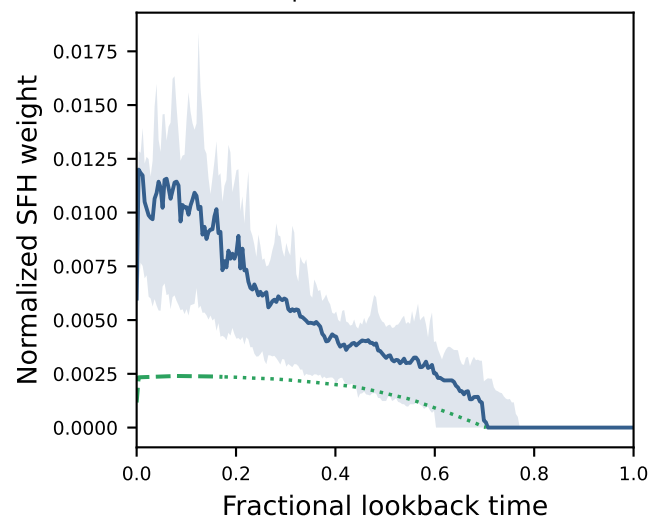
Complete candidate SFH



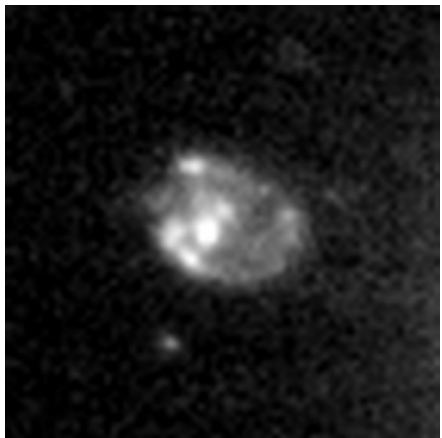
rank 4: 2701411071638079654
 $\log M^*=9.96 (-0.03)$ $z=0.724$
 PHZ $\Delta MS=+1.06$; SFH-100Myr $\Delta MS=+0.70$
 $D=0.025$



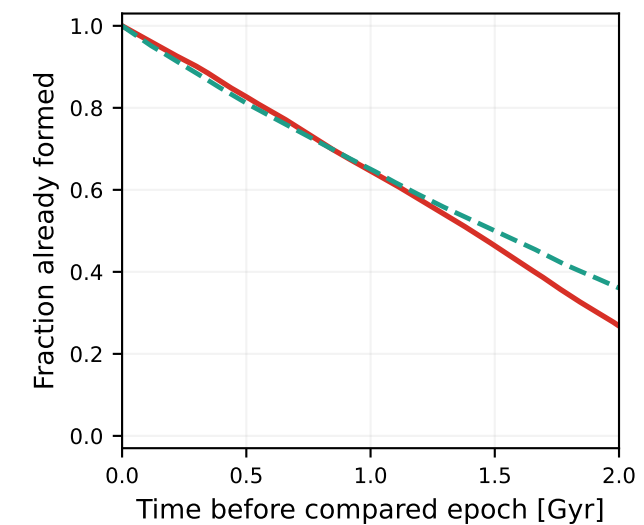
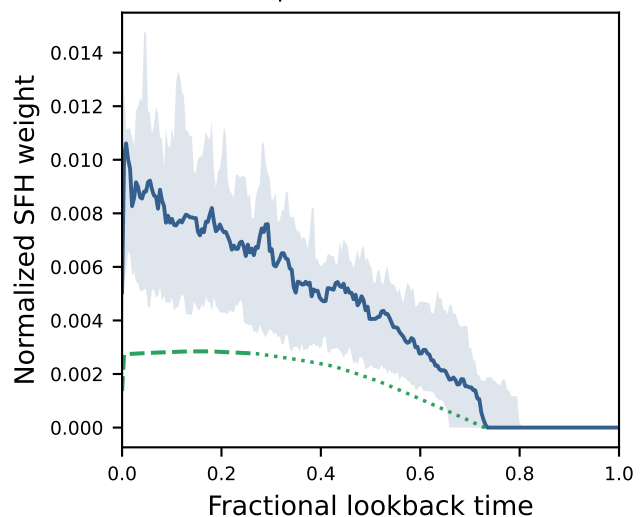
Complete candidate SFH



rank 5: 2714302483670486073
 $\log M^*=10.03 (+0.03)$ $z=0.950$
 PHZ $\Delta MS=+1.59$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.025$



Complete candidate SFH

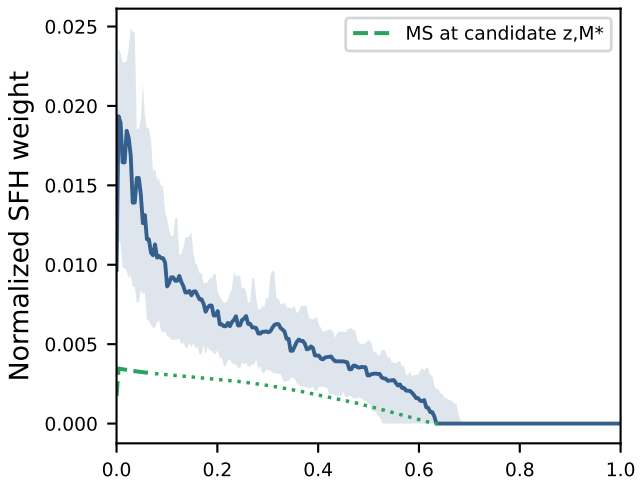


Earlier state $f=0.40$: lookback=3.81 Gyr, predicted $\log M^*=9.82$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

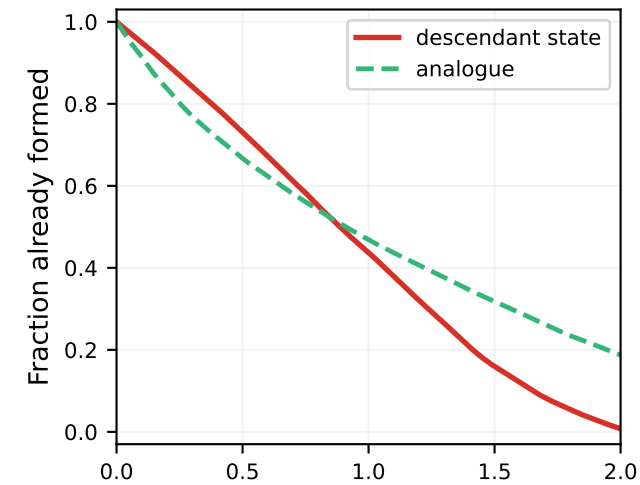
rank 1: 2766281064676170516
 $\log M^*=9.85 (+0.02)$ $z=1.138$
 PHZ $\Delta MS=+1.74$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.090$



Complete candidate SFH

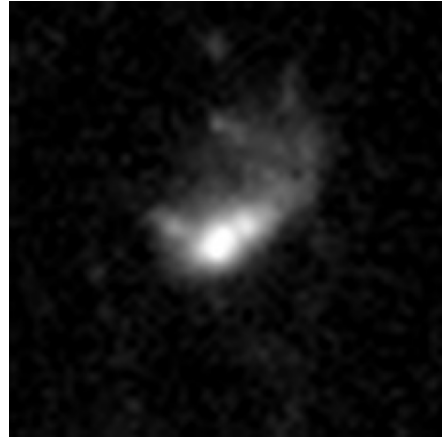


Fractional lookback time

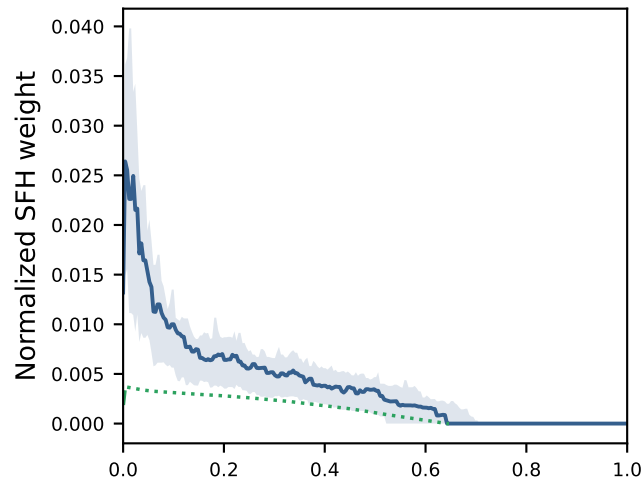


Time before compared epoch [Gyr]

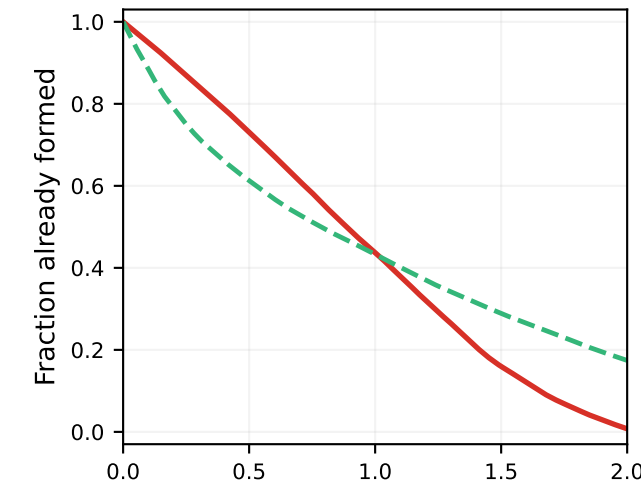
rank 2: 2645606478643808588
 $\log M^*=9.72 (-0.10)$ $z=1.116$
 PHZ $\Delta MS=+1.67$; SFH-100Myr $\Delta MS=+0.83$
 $D=0.096$



Complete candidate SFH

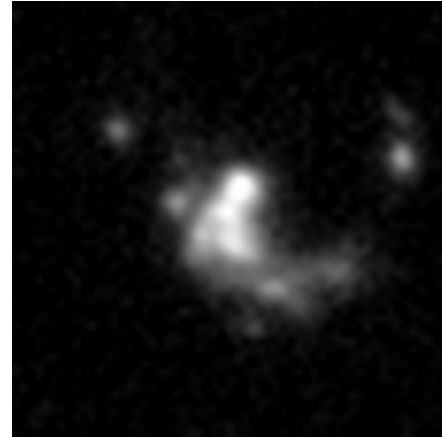


Fractional lookback time

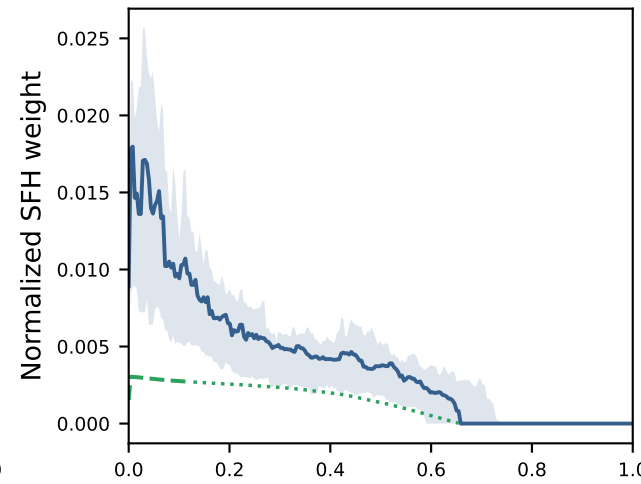


Time before compared epoch [Gyr]

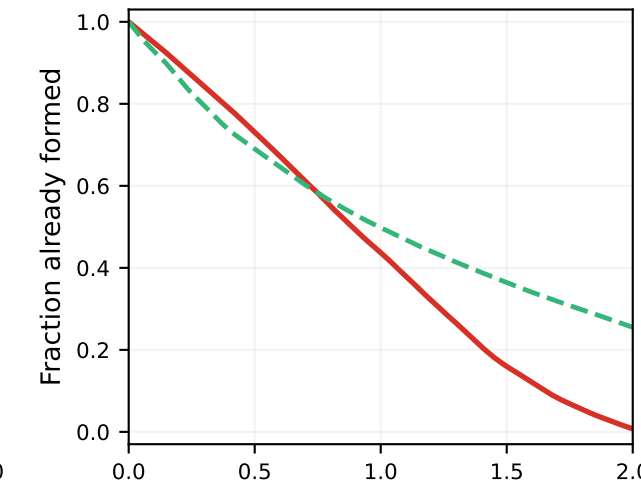
rank 3: 2650234495658951302
 $\log M^*=9.93 (+0.11)$ $z=1.010$
 PHZ $\Delta MS=+1.36$; SFH-100Myr $\Delta MS=+0.74$
 $D=0.108$



Complete candidate SFH

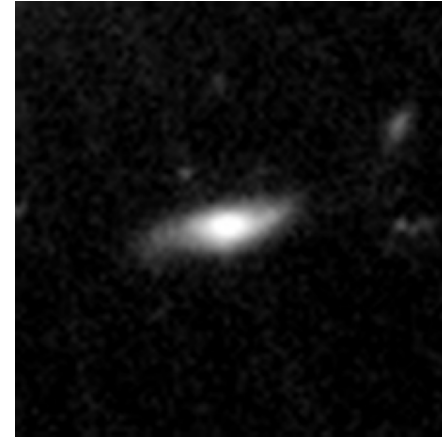


Fractional lookback time

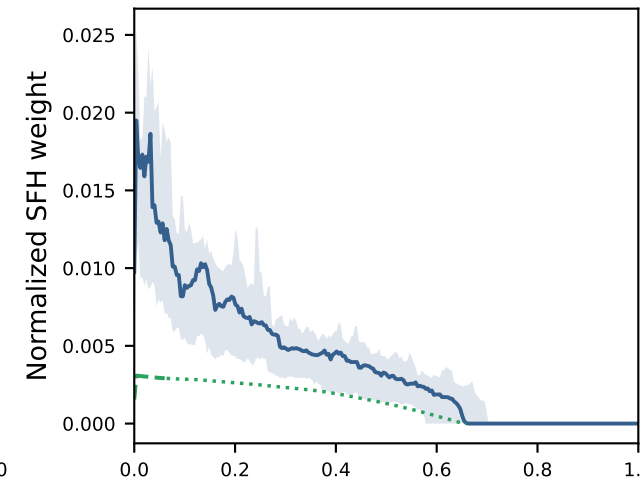


Time before compared epoch [Gyr]

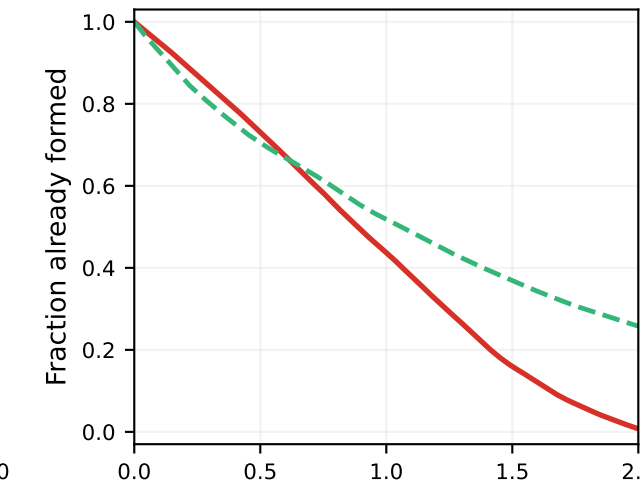
rank 4: 2708364677681344616
 $\log M^*=9.82 (+0.00)$ $z=0.920$
 PHZ $\Delta MS=+1.38$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.113$



Complete candidate SFH

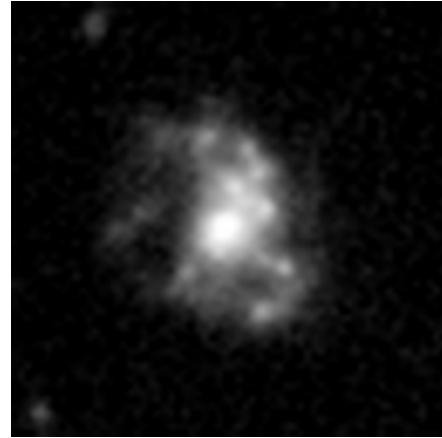


Fractional lookback time

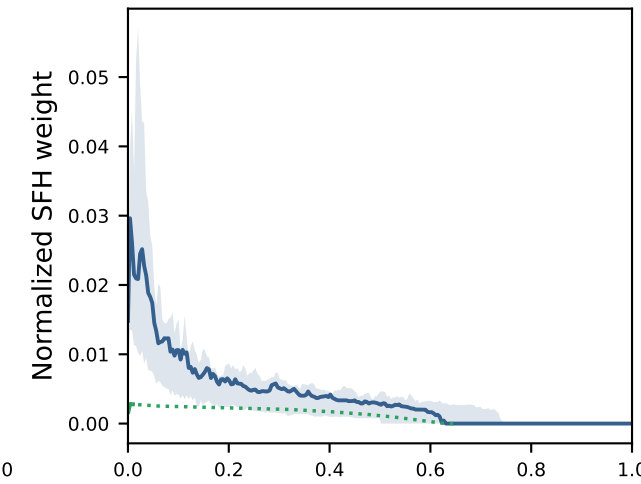


Time before compared epoch [Gyr]

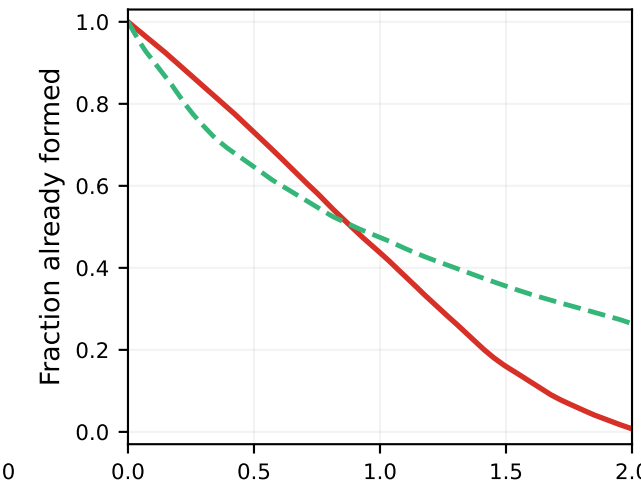
rank 5: 2733550546657853038
 $\log M^*=9.76 (-0.07)$ $z=0.760$
 PHZ $\Delta MS=+1.25$; SFH-100Myr $\Delta MS=+0.7$
 $D=0.117$



Complete candidate SFH



Fractional lookback time



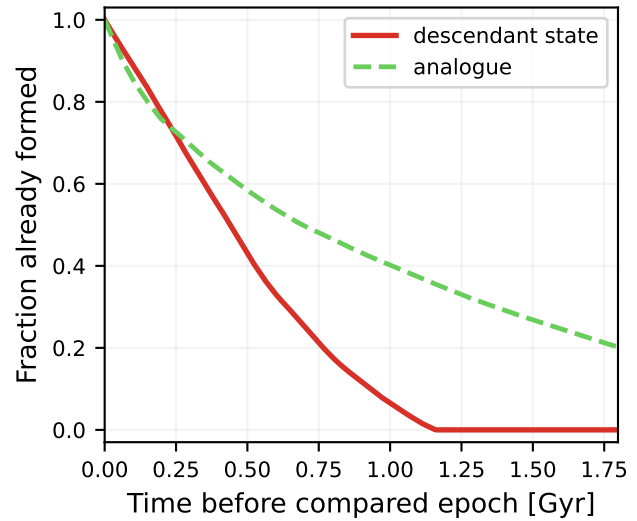
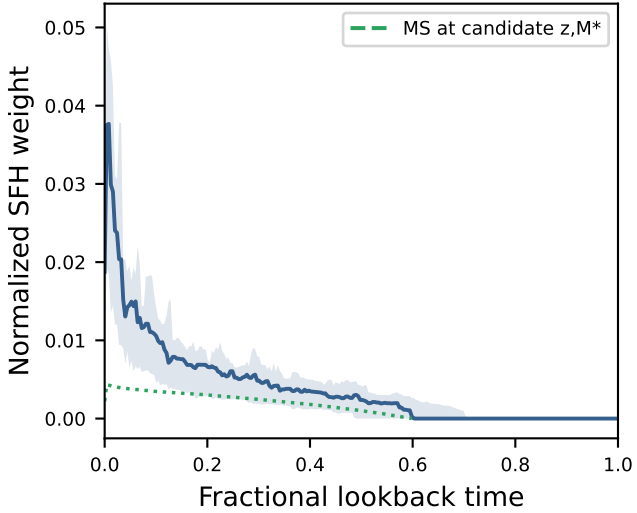
Time before compared epoch [Gyr]

Earlier state $f=0.20$: lookback=4.69 Gyr, predicted $\log M^*=9.52$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

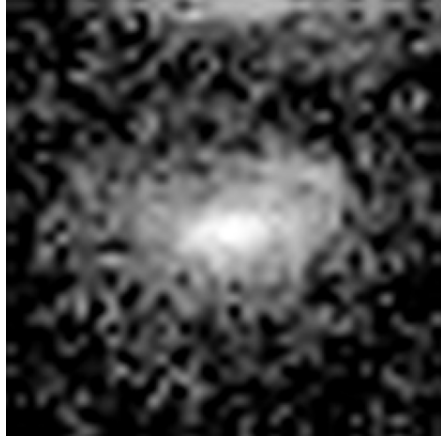
rank 1: 2648545772681764910
 $\log M^*=9.39$ (-0.13) $z=0.994$
 PHZ $\Delta MS=+1.79$; SFH-100Myr $\Delta MS=+0.90$
 $D=0.215$



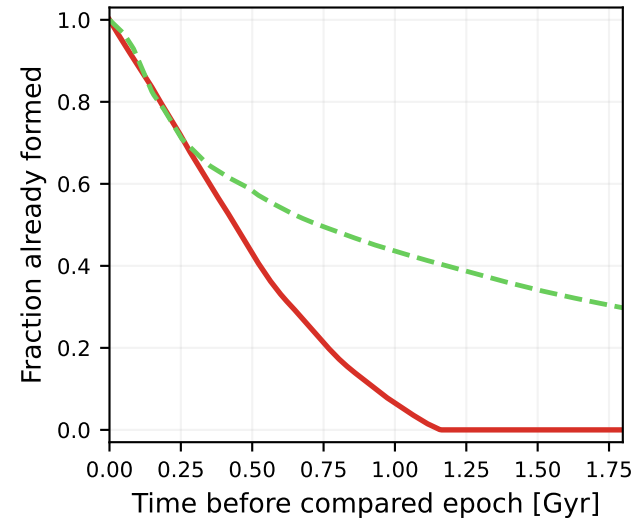
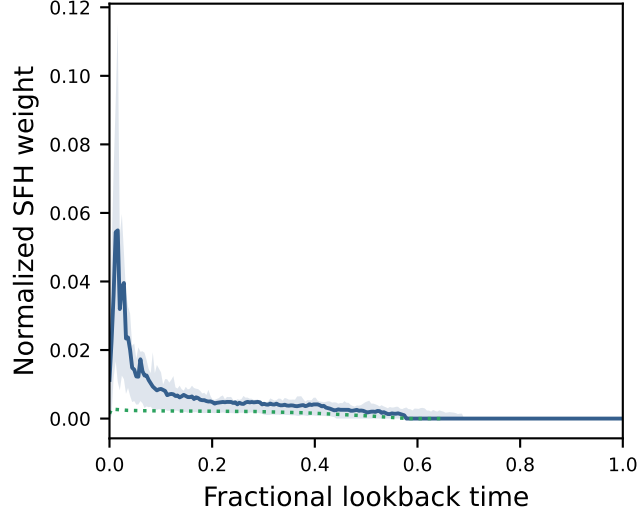
Complete candidate SFH



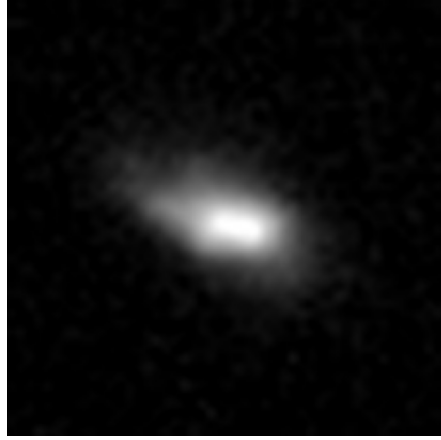
rank 2: 2647975483685285807
 $\log M^*=9.44$ (-0.09) $z=0.510$
 PHZ $\Delta MS=+1.34$; SFH-100Myr $\Delta MS=+1.07$
 $D=0.245$



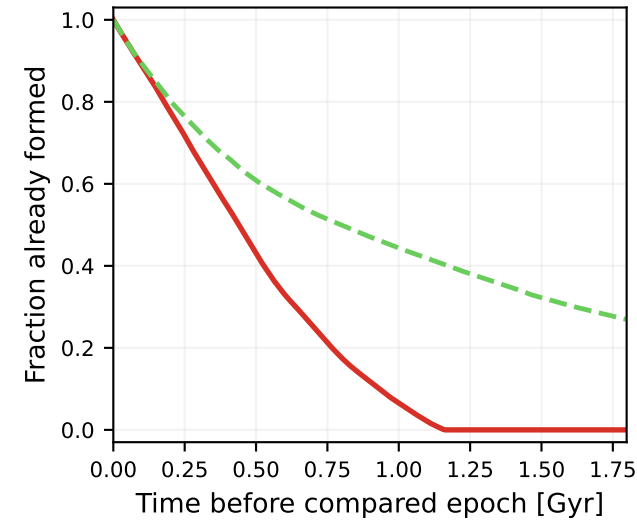
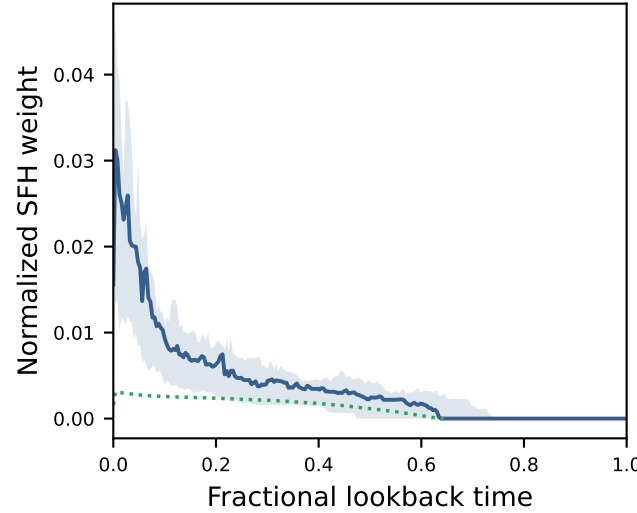
Complete candidate SFH



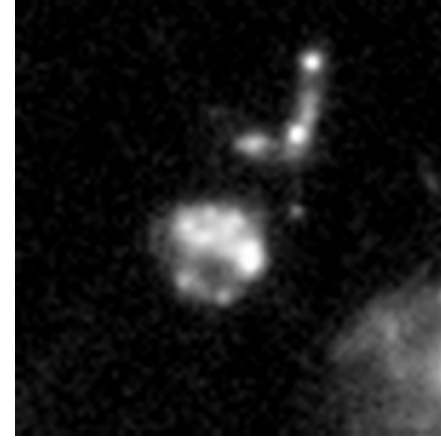
rank 3: 2744318378668883710
 $\log M^*=9.65$ (+0.12) $z=0.770$
 PHZ $\Delta MS=+1.31$; SFH-100Myr $\Delta MS=+0.98$
 $D=0.251$



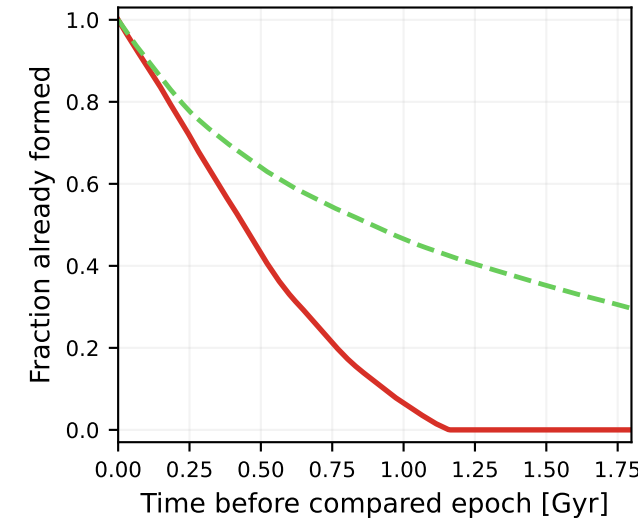
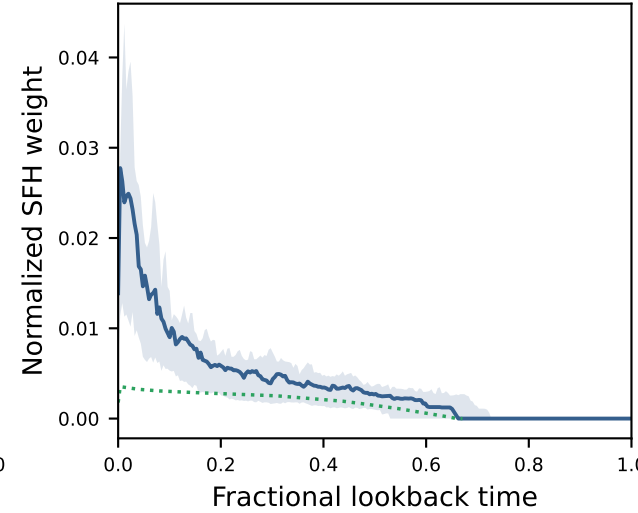
Complete candidate SFH



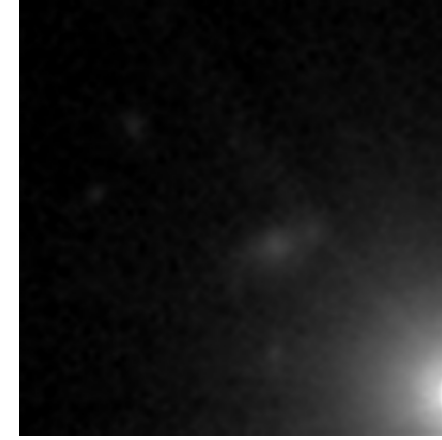
rank 4: 2740268308665100250
 $\log M^*=9.47$ (-0.05) $z=0.760$
 PHZ $\Delta MS=+1.77$; SFH-100Myr $\Delta MS=+0.87$
 $D=0.275$



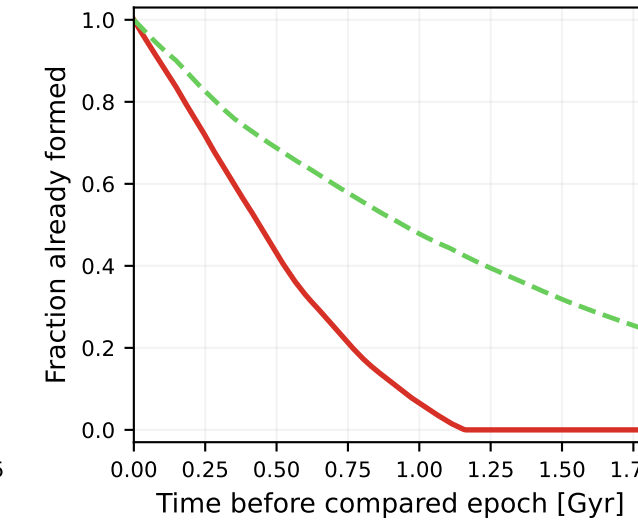
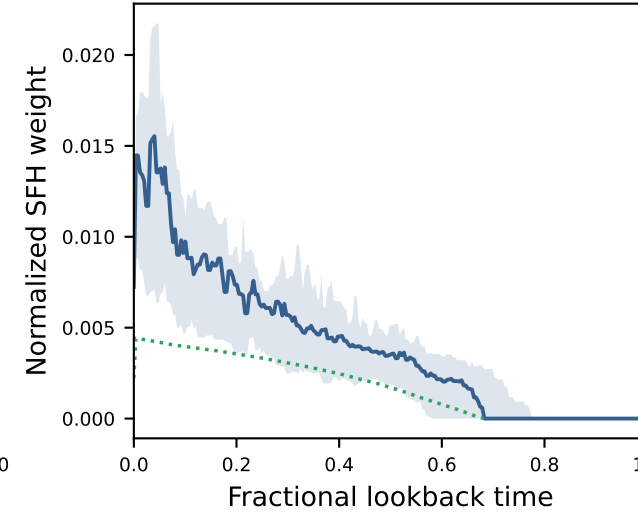
Complete candidate SFH



rank 5: 2695901997638798783
 $\log M^*=9.55$ (+0.03) $z=1.234$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=+0.5$
 $D=0.284$



Complete candidate SFH

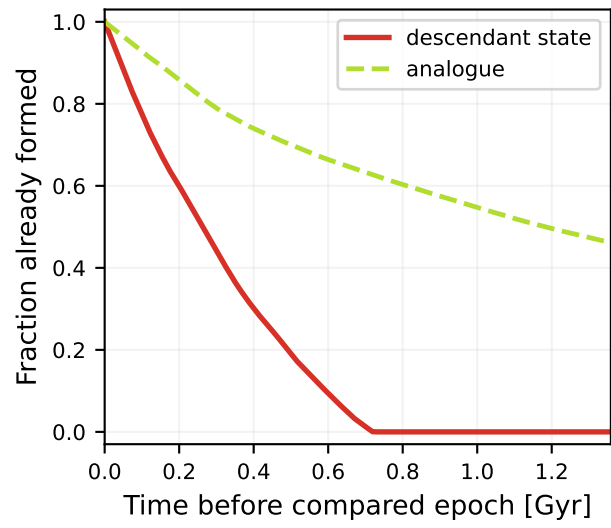
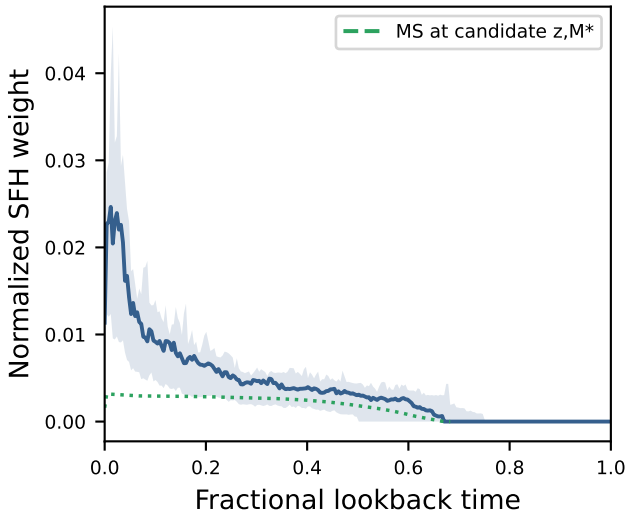


Earlier state $f=0.10$: lookback=5.13 Gyr, predicted $\log M^*=9.22$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

rank 1: 2657573134652992420
 $\log M^*=9.35 (+0.13)$ $z=0.560$
 PHZ $\Delta MS=+1.13$; SFH-100Myr $\Delta MS=+0.86$
 $D=0.453$



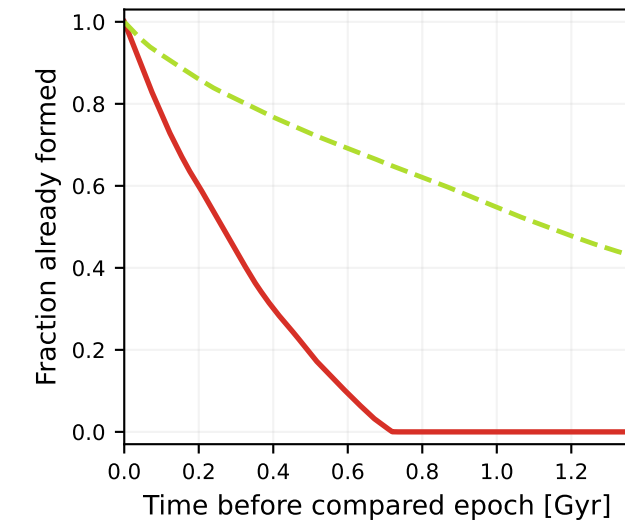
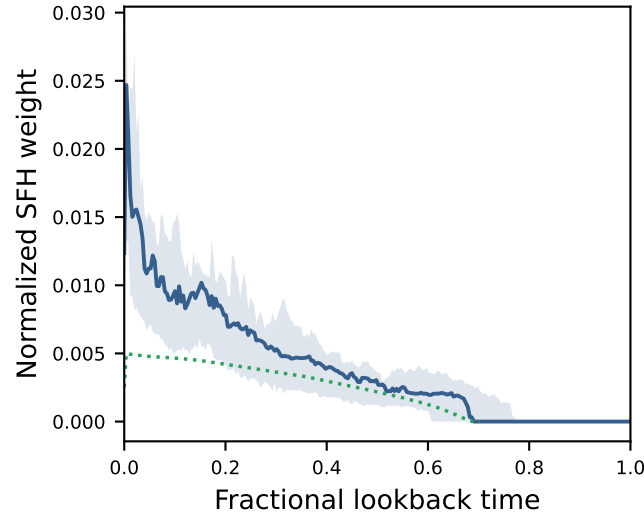
Complete candidate SFH



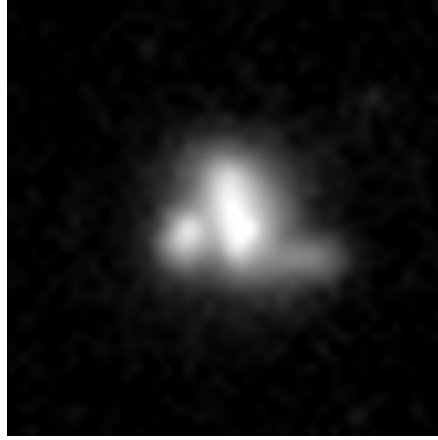
rank 2: 2674598393649348048
 $\log M^*=9.10 (-0.12)$ $z=0.848$
 PHZ $\Delta MS=+1.80$; SFH-100Myr $\Delta MS=+0.62$
 $D=0.460$



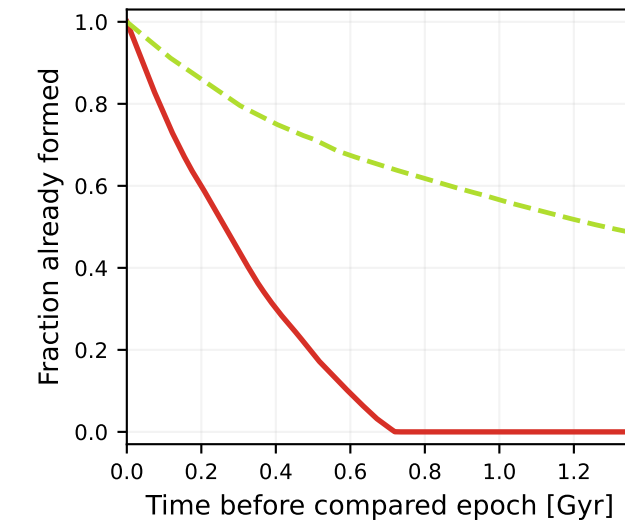
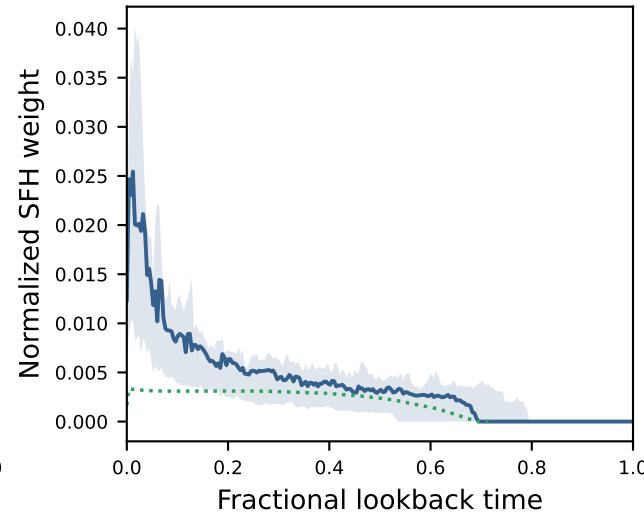
Complete candidate SFH



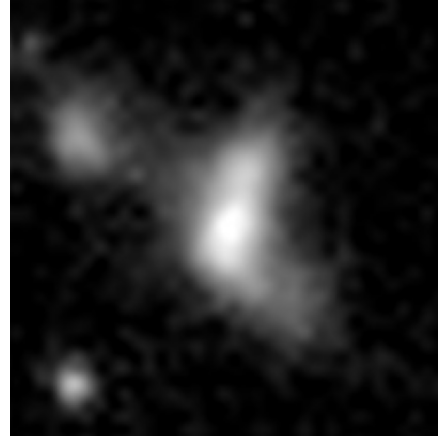
rank 3: 2660165531665312424
 $\log M^*=9.30 (+0.08)$ $z=0.560$
 PHZ $\Delta MS=+1.74$; SFH-100Myr $\Delta MS=+0.87$
 $D=0.466$



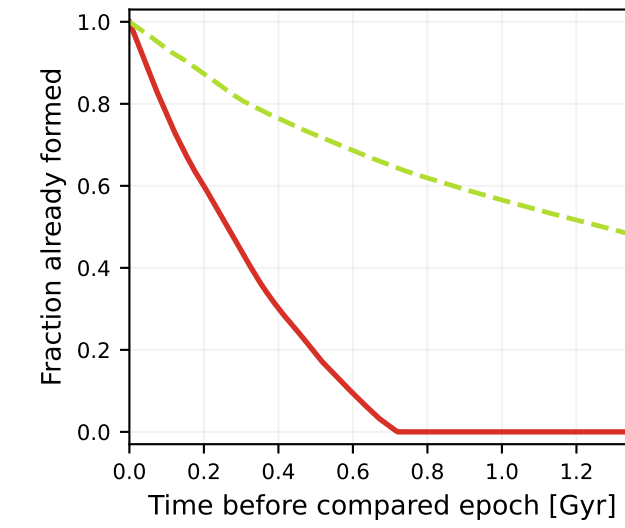
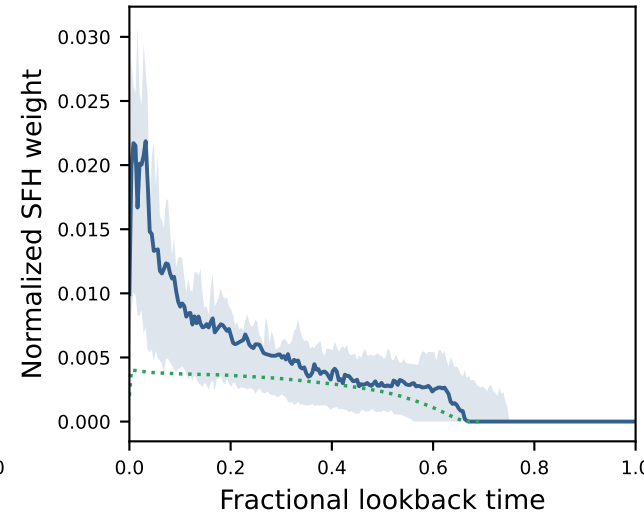
Complete candidate SFH



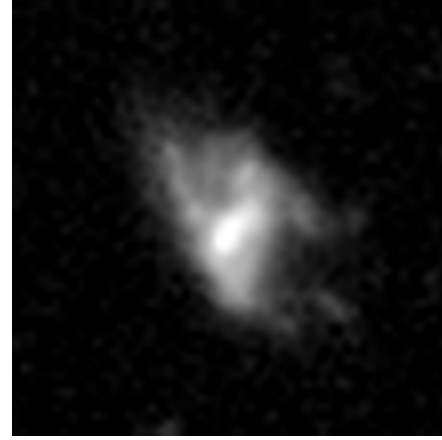
rank 4: 2765951524658897054
 $\log M^*=9.10 (-0.12)$ $z=0.580$
 PHZ $\Delta MS=+1.72$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.471$



Complete candidate SFH



rank 5: 2631142758659540343
 $\log M^*=9.30 (+0.08)$ $z=0.560$
 PHZ $\Delta MS=+1.03$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.505$



Complete candidate SFH

